# The briefing: evidence, inference, prediction

**LEARNING TARGET · R1** I can separate evidence from inference and name the measurements needed to predict a future position.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Evidence** | What was actually stated or measured. "An object is approaching." |
| **Inference** | A conclusion that still needs testing. "It will hit Earth." |
| **Model** | A simplified description with stated assumptions that makes a testable prediction. |
| **Testable question** | "Is it bad?" → "Will its predicted path intersect Earth's position at the same time?" |

## SECTION B MODEL + WHAT A PREDICTION NEEDS

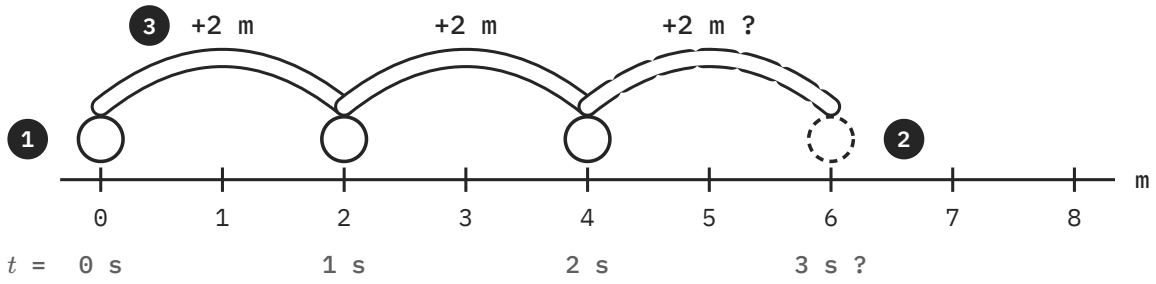
$$\text{new position} = \text{starting position} + \text{change} \times \text{elapsed time}$$

**needs**
- current position
- velocity *with direction*
- elapsed time + does $v$ change?

CLAIM · EVIDENCE · REASONING · LIMIT

## SECTION C COLOR IT · WATCH A PATTERN FORM

*One dot marks the cart's position each second.*




- ☐ **1 · BLUE** Observations: measured positions
- ☐ **2 · BLUE** Prediction: dashed outline
- ☐ **3 · PURPLE** The pattern: +2 m each second

## SECTION D WORKED EXAMPLE

A cart is at 10 m, 8 m, and 6 m at 0, 1, and 2 s. Predict its position at 4 s, and name an observation that would challenge the model.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $x$ = 10, 8, 6 m<br>$t$ = 0, 1, 2 s | new $x$ = start + change × time | change = −2 m each second<br>$x$(4 s) = 6 + (−2)(2) | **2 m**<br>limit: a 3 m reading would challenge it |

## SECTION E THE 2026-XJ THREAD

DAY 1 OF 22 · NEW TOOL: EVIDENCE VS. INFERENCE

The desk holds evidence (what was measured) and inference (what it suggests). A forecast for 2026-XJ must rest on the evidence: positions, times, and their uncertainty.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# What does the evidence let us say?

Evidence, inference, models, and prediction

Suppose I tell you that an object in space is approaching Earth. It is tempting to jump straight to a collision. But approaching Earth and hitting Earth are different claims. Before we predict what will happen, we need to separate what we know from what we think it means. That is where our work in physics begins.

Use evidence and inference in **Section A** to separate two claims. Evidence is what was observed, measured, or stated in the information we were given. An inference is a conclusion we draw from that evidence. An inference can be reasonable and still need testing. If we have a series of measured positions, those positions are evidence. Saying the object will continue along the same path is an inference that depends on an assumption about its motion.

A model makes that assumption clear. It is a simplified description that helps us explain a pattern and make a prediction we can test. Imagine a cart at **0 m**, then **2 m**, then **4 m**, with one second between measurements. Its position increases by **2 m each second**. If that pattern continues, we predict **6 m at 3 s**. The prediction is not another measurement. In **Section C**, compare the dashed prediction with the measured dots. Blue identifies position; purple identifies the velocity pattern connecting the positions.

To predict a later position, we need a known starting position, a velocity with direction, and the time elapsed from that starting measurement. We also need to decide whether treating the velocity as constant is reasonable. With constant velocity, the model is $\boldsymbol{x_f = x_i + v\Delta t}$: final position equals initial position plus velocity times elapsed time. The velocity tells us how much position changes each second. Multiplying meters per second by seconds gives a change in meters.

Now test the pattern with the cart in **Section D**. Its positions are **10 m, 8 m, and 6 m** at **0 s, 1 s, and 2 s**. Position decreases by **2 m each second**, so its velocity is **−2 m/s**. From the last measurement at 2 s to the requested time of 4 s, another 2 s pass. The predicted position is **6 m + (−2 m/s)(2 s) = 2 m**. The negative sign tells us that position is decreasing along our chosen axis.

I want the reasoning and its limit to travel with that answer. Our claim is 2 m at 4 s. Our evidence is the measured pattern. Our reasoning extends that pattern for two more seconds. Our limit is the assumption that the velocity stays constant. A new reading of 3 m at 4 s would make us check the measurements and the model, including whether measurement uncertainty explains the difference or the motion changed.

This is how we will investigate our fictional asteroid, 2026-XJ. A useful question is whether its predicted path reaches Earth's position at the same time Earth is there. We will build that prediction from measurements, explain our assumptions, and revise our conclusion when the evidence calls for it.

# Where are you? How far did you travel?

**LEARNING TARGET · K1** I can describe a route using an origin, signed position, total distance, and displacement.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Position** | A signed coordinate in a reference frame: choose an origin and a + direction. $x$ = −2 m is 2 m left of the origin. |
| **Distance** | Sum of the length of every leg: $\lvert x_2 - x_1\rvert + \lvert x_3 - x_2\rvert + \ldots$ Never negative. |
| **Displacement** | $\Delta x = x_f - x_i$. Its sign depends on the axis you chose; the physical motion does not. |
| **Scalar / vector** | A scalar has magnitude + units (distance, time). A vector also has direction (displacement). |

## SECTION B KEY EQUATIONS + UNITS

$$\Delta x = x_f - x_i$$

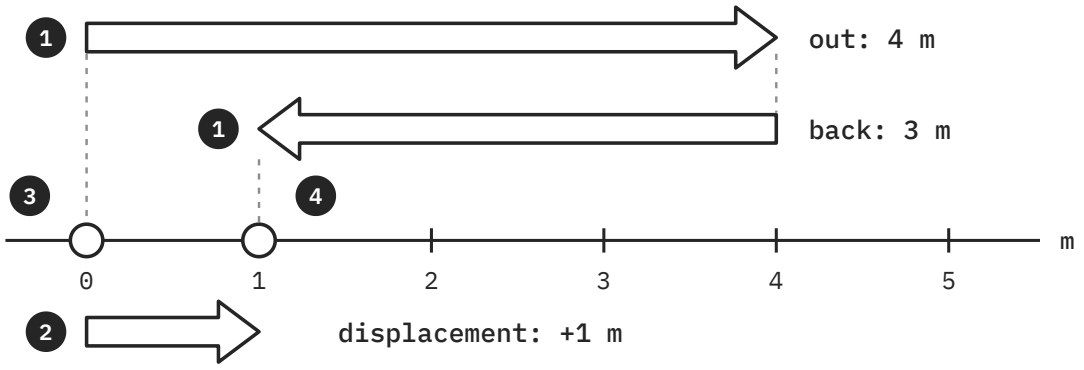
$$\text{distance} = \Sigma\ \lvert\text{each leg}\rvert$$

$\Delta$ "change in" (final − initial)

$x$ signed position, meters (m)

## SECTION C COLOR IT · ONE TRIP, THREE QUESTIONS

*0 → 4 → 1 m: |4 − 0| + |1 − 4| = 7 m; Δx = +1 m*




- ☐ **1 · PENCIL** Distance: trace both legs
- ☐ **2 · BLUE** Displacement arrow
- ☐ **3 · BLUE** Start position
- ☐ **4 · BLUE** Finish position

## SECTION D WORKED EXAMPLE

Start at −3 m, move to +2 m, then finish at −1 m. Find the distance and the displacement, with units.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $x_i$ = −3 m<br>turn at +2 m<br>$x_f$ = −1 m | distance = $\Sigma\ \lvert\text{leg}\rvert$<br>$\Delta x = x_f - x_i$ | $\lvert 2 - (-3)\rvert + \lvert -1 - 2\rvert = 5$<br>+ 3<br>−1 − (−3) | distance **8 m**<br>displacement **+2 m** |

## SECTION E THE 2026-XJ THREAD

DAY 2 OF 22 · NEW TOOL: A REFERENCE FRAME

XJ's position only means something in a reference frame: an origin, axes, and a sign convention. Its displacement, not its path length, decides where it ends up.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# One trip can have several correct descriptions

Position, distance, displacement, and reference frames

If I ask where you are, an answer like “two meters” leaves out something important: two meters from where? A position needs a reference frame. For motion along a straight line, we choose an origin, label it zero, and choose a positive direction. Once we agree on those choices, a position gives us a location we can interpret.

Suppose right is positive. A position of $x = -2$ **m** means the object is two meters left of the origin. That sign describes location. It does not tell us which way the object is moving. An object at −2 m could move right, move left, or remain there. Keep location and motion separate as you read a situation.

Trace the two legs in **Section C**, starting at **0 m**, traveling right to **4 m**, and returning to **1 m**. The distance traveled counts the entire route. The outward leg is 4 m and the return leg is 3 m, giving **7 m** altogether. Returning does not erase distance already traveled. Distance is never negative because each leg contributes its length.

Use the displacement equation in **Section B** to ask a different question: how did position change from the start to the finish? The symbol $\Delta$ means “change in,” so $\Delta x$ means change in position. We calculate final minus initial: $\Delta x = x_f - x_i$. For this trip, that is **1 m − 0 m = +1 m**. The object finishes one meter to the right of where it began. The 7 m distance and +1 m displacement both describe this same trip.

This is also the difference between a scalar and a vector. A scalar has a magnitude and units; distance and elapsed time are examples. A vector also has direction. Displacement is a vector, and on our one-dimensional axis its sign communicates direction. In the diagram, the traveled path stays in pencil. The starting and finishing positions and the displacement arrow are blue because they describe position and its change.

The trip in **Section D** begins at **−3 m**, turns at **+2 m**, and finishes at **−1 m**. The first leg has length $|2 - (-3)| = 5$ **m**. The second has length $|-1 - 2| = 3$ **m**. The bars mean absolute value: we use the size of each position change when adding distance. Total distance is therefore **8 m**. Displacement uses only the endpoints: **−1 m − (−3 m) = +2 m**.

Notice that the final position is negative while the displacement is positive. The object finishes left of the origin, but right of its starting position. If we reversed which direction we called positive, the coordinate signs would reverse. The physical trip and its distance would stay the same. I want your signs to communicate the reference frame you chose.

For our fictional asteroid, 2026-XJ, we will need a clear origin, axes, and sign convention before a predicted position has meaning. Its total distance traveled tells us about the route. Its starting position and displacement tell us where it finishes. That distinction matters when we compare its predicted location with Earth's.

# Velocity and position–time graphs

**LEARNING TARGET · S1** I can calculate average velocity and interpret a position–time graph using slope.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Average velocity** | $\bar{v} = \Delta x \div \Delta t$. Carries direction; summarizes a whole interval. |
| **Average speed** | Total distance ÷ elapsed time. No direction. |
| **Slope** | On an $x$–$t$ graph, slope = velocity (units m/s). Height = position, not velocity. |
| **Instantaneous velocity** | Velocity at one moment: the tangent slope on a curved $x$–$t$ graph. |

## SECTION B KEY EQUATION + UNITS

$$\bar{v} = \Delta x \div \Delta t$$

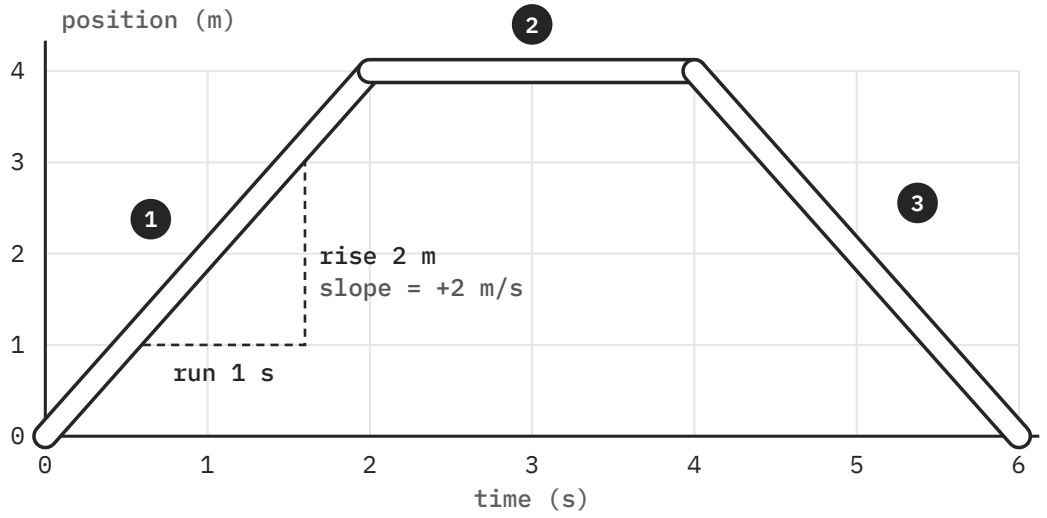
$$\Delta x = x_f - x_i$$

$v$ velocity, meters per second (m/s)

speed total distance ÷ $\Delta t$

## SECTION C COLOR IT · READ A GRAPH AS A MOTION STORY

*Height = position. Slope = velocity.*




☐ **1 · PURPLE** Rising: moving +, $v > 0$

☐ **2 · PURPLE** Flat: stopped, $v = 0$

☐ **3 · PURPLE** Falling: moving –, $v < 0$

## SECTION D WORKED EXAMPLE

A straight $x$–$t$ line passes through (2 s, 14 m) and (6 s, 6 m). Find the velocity, then predict $x$ at 8 s.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| (2 s, 14 m)<br>(6 s, 6 m)<br>predict at 8 s | $v = \Delta x \div \Delta t$<br>$x = x_i + v\,t$ | $v = (6 - 14) \div (6 - 2)$<br>$x(8) = 6 + (-2)(2)$ | **–2 m/s; 2 m**<br>if the line continues |

## SECTION E THE 2026-XJ THREAD

DAY 3 OF 22 · NEW TOOL: AVERAGE VELOCITY

Average velocity, $\Delta x \div \Delta t$ from two tracked positions, is the desk's first estimate of how fast XJ closes in. The slope of its $x$–$t$ graph shows whether that rate holds.

*Unit question: How fast is it moving, and where will it arrive?*

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# A graph can tell the story of motion

Average velocity, speed, and position-time graphs

We can now describe where an object starts and finishes. The next question is how quickly its position changes. Moving four meters in two seconds and moving four meters in twenty seconds produce the same displacement, but they describe different motion. Time lets us tell the difference.

Use the ratio in **Section B** for average velocity: $\bar{v} = \Delta x \div \Delta t$. The bar means average over an interval. We use final minus initial for both changes. Velocity includes direction, so its sign depends on the axis we chose. Average speed uses total distance divided by elapsed time. Speed is nonnegative. When an object changes direction, average speed and the magnitude of average velocity can be different.

A position-time graph puts time on the horizontal axis and position on the vertical axis. Every point identifies where the object is at a particular time. The graph's height gives position. Its slope, the vertical change divided by the horizontal change, gives velocity. The units show why: a change in meters divided by a change in seconds produces meters per second. This graph is a record of position over time, not a picture of the road's shape.

Trace the three segments of the position-time graph in **Section C**. From **0 s to 2 s**, position rises from **0 m to 4 m**. The slope is **+4 m ÷ 2 s = +2 m/s**. From 2 s to 4 s, position remains at 4 m. The horizontal line has zero slope, so the object is stopped. From 4 s to 6 s, it returns from 4 m to 0 m. The slope is **−4 m ÷ 2 s = −2 m/s**. It moves in the negative direction at a speed of 2 m/s. The purple marks connect each slope to its velocity.

Across that entire six-second trip, the object returns to its starting position. Its displacement is zero, so its average velocity is **0 m/s**. It still travels 4 m out and 4 m back. Its average speed is **8 m ÷ 6 s ≈ 1.33 m/s**. An average velocity of zero over the whole trip does not mean the object was stopped throughout it.

On a straight segment, the slope stays constant. A steeper segment means a larger speed because more position change occurs in the same time. A curved graph has a changing slope. Average velocity between two points uses the slope of the line joining them. Instantaneous velocity, the velocity at one moment, uses the slope of a tangent: a line that matches the curve's direction at that point.

In **Section D**, the straight line passes through **(2 s, 14 m)** and **(6 s, 6 m)**. Velocity is **(6 − 14) m ÷ (6 − 2) s = −2 m/s**. If that velocity continues, two more seconds bring another −4 m of displacement. At 8 s the predicted position is **6 m + (−2 m/s)(2 s) = 2 m**. Extending the straight line makes the same prediction.

For our fictional asteroid, 2026-XJ, tracked positions and times let us estimate velocity along a chosen axis. I want the graph and calculation to tell the same story. We can then judge whether continuing with a constant-velocity prediction is reasonable or whether changing slopes call for a different model.

# Velocity graphs and components

**LEARNING TARGET · R2** I can read $v$–$t$ areas and resolve velocity into components.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Signed area** | Above zero adds displacement; below zero subtracts it. |
| **Distance** | Add area sizes, including the return trip. |
| **Component** | A vector's signed part along one axis. |
| **Angle $\theta$** | Name the starting axis and turn direction. |

## SECTION B KEY EQUATIONS + UNITS

$$\Delta x = \text{signed area}$$

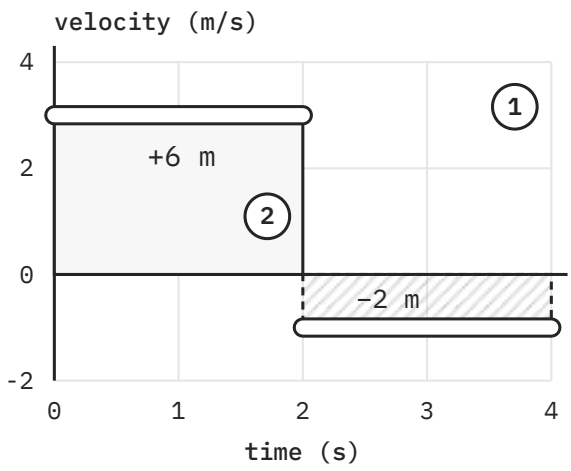
Split $v$ along fixed axes

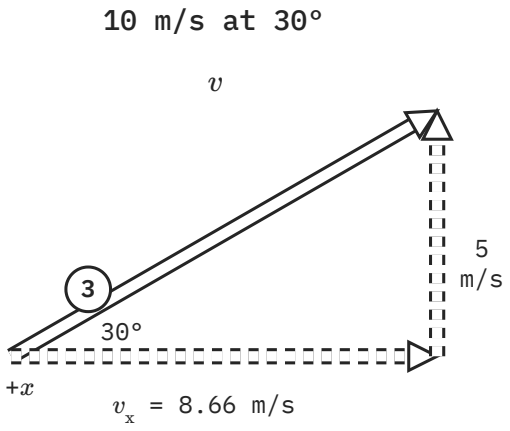
$v_x$ $v \cos\theta$

$v_y$ $v \sin\theta$; $\theta$ from $+x$

## SECTION C COLOR IT · AREA AND COMPONENTS

*Two views of motion.*




☐ **1 · PURPLE** Velocity and its parts

☐ **2 · BLUE** Signed area: displacement

☐ **3 · PENCIL** Axes and angle

## SECTION D WORKED EXAMPLE

A cart moves at **+3 m/s** for **2 s**, then **−1 m/s** for **2 s**. **Find its displacement and total distance.**

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $v_1$ = **+3 m/s**<br>$v_2$ = **−1 m/s**<br>**2 s** per interval | Use areas because each interval has steady $v$. | $\Delta x$ = 3(2) − 1(2)<br>distance = 6 + 2 | **+4 m; 8 m**<br>Return travel reduces $\Delta x$, not distance. |

## SECTION E THE 2026-XJ THREAD

DAY 4 OF 22 · NEW TOOL: SIGNED AREA

For fictional XJ, the area of a velocity–time graph gives a change in position. Components separate motion toward Earth from motion across its line of sight.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Velocity tells us more than speed

Signed area, displacement, and velocity components

A velocity-time graph lets us recover a trip even when no positions are listed. The height tells us velocity at a particular time. To find displacement, we need to combine that velocity with how long it lasts. I want you to keep those two jobs separate: reading a height gives meters per second; finding an area gives meters.

For constant velocity, $\Delta x = v\Delta t$. On the graph, that multiplication is the area of a rectangle: height times width. Above the time axis, velocity is positive and the area adds positive displacement. Below it, velocity is negative and the area adds negative displacement. This is signed area. The blue areas on your notes page represent changes in position, even though the vertical axis describes velocity.

Trace the two signed areas on the left of **Section C**. The cart travels at **+3 m/s for 2 s**, then **−1 m/s for 2 s**. Its first displacement is **(+3 m/s)(2 s) = +6 m**. Its second is **(−1 m/s)(2 s) = −2 m**. Combining them gives $\Delta x$ = **+6 m − 2 m = +4 m**. The cart finishes four meters in the positive direction from where it started. That does not mean it traveled only four meters.

Check the two answers in **Section D**. Distance counts both legs by their lengths, so **distance = 6 m + 2 m = 8 m**. When a velocity graph crosses zero, split the areas there before adding their sizes for distance. A negative region does not undo travel already completed. A horizontal segment below zero also describes steady motion, not an object slowing down. Its velocity stays negative and its speed stays constant.

Velocity can also point at an angle. Then we describe the same motion using two perpendicular components, one along each axis. A component is a signed part of a vector. The object is not making two separate trips. Both components happen during the same interval and together describe the single velocity arrow.

On the right of **Section C**, the velocity has magnitude **10 m/s** and points **30° from the positive horizontal axis toward the positive vertical axis**. With that angle measured from the horizontal, $v_x = v\cos\theta$ and $v_y = v\sin\theta$. Therefore $v_x$ = **(10 m/s) cos 30° ≈ 8.66 m/s** and $v_y$ = **(10 m/s) sin 30° = 5.00 m/s**. Both signs are positive because the arrow points toward increasing coordinates on both axes.

The angle's reference axis matters. Before choosing sine or cosine, identify which component is adjacent to that angle and which is opposite. Check the sketch as well as the calculator. Neither component should exceed the magnitude of the original vector. Purple identifies all three velocity arrows; the components explain the direction of the original arrow.

For our fictional asteroid, 2026-XJ, motion toward Earth and motion across our line of sight both matter. One component alone cannot describe the entire path. Components let us predict each coordinate, while signed area keeps track of how motion along each axis changes position.

# Position predictions and their limits

**LEARNING TARGET · R3** I can predict a position and state its uncertainty.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Start** $x_0$ | Position when the chosen interval begins. |
| **Bound** | An allowed range, not a standard deviation. |
| **Half-width** | The ± value; full width is twice as large. |
| **Assumption** | A condition treated as true, such as steady $v$. |

## SECTION B KEY EQUATIONS + UNITS

$$x_f = x_0 + vt$$

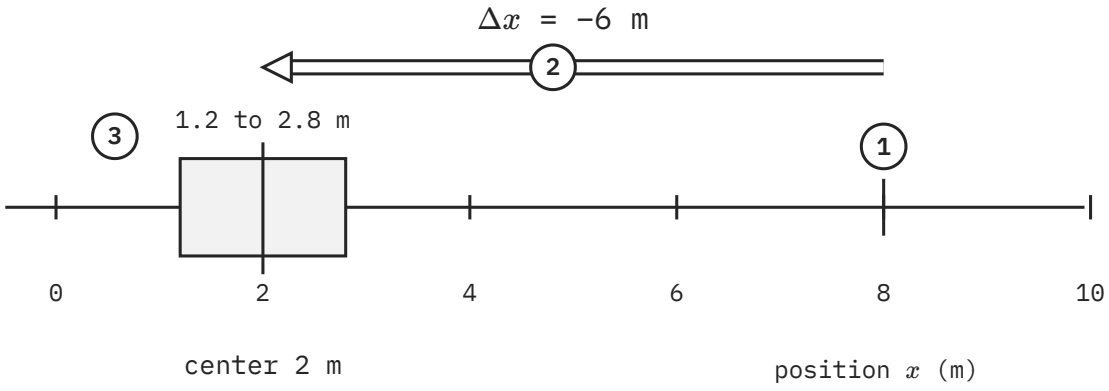
$$\delta x_f = \delta x_0 + t\delta v$$

$t$ exact time, s
$\delta x$ bound, m

## SECTION C COLOR IT · A BOUNDED FORECAST

*Center plus range.*

$x_0 = 8 \pm 0.5$ m $\qquad$ $v = -2.0 \pm 0.1$ m/s; $t = 3$ s




☐ **1 · BLUE** Start and end positions

☐ **2 · BLUE** Displacement

☐ **3 · BLUE** Prediction interval

## SECTION D WORKED EXAMPLE

A cart starts at **8 ± 0.5 m** and moves at **−2.0 ± 0.1 m/s** for an exact **3 s**. **Find its predicted position interval.**

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $x_0$ = **8 ± 0.5 m**<br>$v$ = **−2.0 ± 0.1 m/s**<br>$t$ = **3 s**; steady $v$ | Steady $v$: $x_f = x_0 + vt$.<br>Add input bounds. | $x_f = 8 - 2(3)$<br>$\delta x = .5 + 3(.1)$<br>= **0.8 m** | **1.2 to 2.8 m**<br>**2.0 ± 0.8 m**; full width **1.6 m**. |

## SECTION E THE 2026-XJ THREAD

DAY 5 OF 22 · NEW TOOL: BOUNDED FORECAST

XJ's position and velocity readings have limits. A forecast should show both the central result and the interval allowed by those limits; the interval does not give an impact probability.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# A prediction should carry its uncertainty

Constant velocity, bounds, and a range of possible positions

A prediction written as one number can look more certain than the measurements supporting it. If our starting position is only known within a range, the predicted position should reflect that range. Today I want us to make both parts of a prediction visible: the central value and how far the allowed values could extend around it.

For steady velocity, $x_f = x_0 + vt$, where time is measured from the starting position. Suppose $x_0 = 8.0 \pm 0.5$ m, $v = -2.0 \pm 0.1$ m/s, and the elapsed time is exactly 3 s in our model. The central prediction is $x_f = 8.0\ \text{m} + (-2.0\ \text{m/s})(3\ \text{s}) = 2.0\ \text{m}$. Position decreases because the velocity is negative.

Here the plus-or-minus amounts are bounds: limits on the values we allow. The starting position could lie anywhere from **7.5 m to 8.5 m**, and the velocity from **−2.1 m/s to −1.9 m/s**. A more negative velocity produces a larger decrease in position during those three seconds. This is why keeping the signs matters when we find the endpoints.

Locate the two ends of the prediction interval in **Section C**. The lowest final position combines the lowest starting position with the most negative velocity: $7.5\ \text{m} + (-2.1\ \text{m/s})(3\ \text{s}) = 1.2\ \text{m}$. The highest combines the highest starting position with the least negative velocity: $8.5\ \text{m} + (-1.9\ \text{m/s})(3\ \text{s}) = 2.8\ \text{m}$. Every allowed combination falls between those limits, provided constant velocity and exact elapsed time remain valid assumptions.

Use the bound equation in **Section B** to check the width of that interval. Let $\delta$ mean the uncertainty bound on a quantity. For this model, $\delta x_f = \delta x_0 + t\delta v$. Thus $\delta x_f = 0.5\ \text{m} + (3\ \text{s})(0.1\ \text{m/s}) = 0.8\ \text{m}$. We report $x_f = 2.0 \pm 0.8$ m. The 0.8 m is the distance from the center to either limit. The full interval is **1.6 m wide**.

Notice how velocity uncertainty grows into position uncertainty as time passes. An error of one tenth of a meter each second becomes three tenths of a meter after three seconds. The initial-position bound remains part of the result too. On the notes page, the central marker and surrounding blue interval show these two ideas without pretending that the center is an exact location.

These bounds are not a probability distribution or a statement that a certain percentage of results must fall inside the interval. We have added the largest allowed contributions. If time also had uncertainty, this expression would need revision. If the object accelerates, the constant-velocity model itself may fail, even when our arithmetic is correct.

For our fictional asteroid, 2026-XJ, a useful forecast includes the reference frame, future time, predicted range, and assumptions. New observations can narrow that range or reveal that the motion changed. Reporting uncertainty makes the claim more informative and gives us something specific to compare with the next measurement.

# Add vectors by their components

**LEARNING TARGET · S2** I can add vectors and resolve their signed components.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Component** | Signed part along an axis; keep axes fixed. |
| **Resultant** | Sum: first tail to final tip. |
| **Magnitude** | Nonnegative size from perpendicular parts. |
| **Quadrant** | Both component signs locate the direction. |

## SECTION B KEY EQUATIONS + UNITS

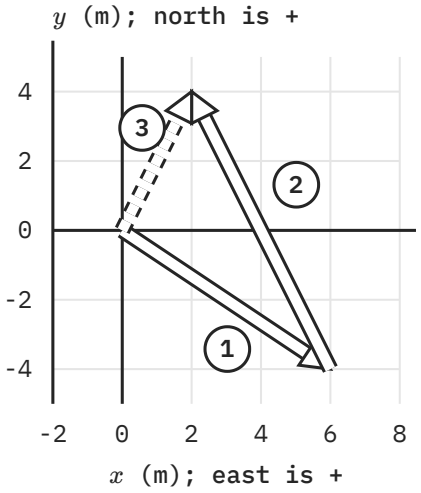
$$R_x = A_x + B_x$$

$$R_y = A_y + B_y$$

$|R|$ $\sqrt{(R_x^2 + R_y^2)}$
angle ratio plus signs

## SECTION C COLOR IT · ADD SIGNED PARTS

*Grid steps: **2 m**.*




$A$ = (6, −4) m

$B$ = (−4, 8) m

$R$ = (2, 4) m

$|R|$ = 4.47 m

☐ **1 · BLUE** Displacement $A$

☐ **2 · BLUE** Displacement $B$

☐ **3 · BLUE** Resultant $R$: dashed

## SECTION D WORKED EXAMPLE

Two displacements are $A$ = (6, −4) m and $B$ = (−4, 8) m. **Find their sum's magnitude and direction.**

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| +$x$ is *east*<br>+$y$ is *north*<br>$A$ and $B$ use these axes. | Add $x$ and $y$.<br>Use the right triangle. | $R$ = (2, 4) m<br>$\|R\|$ = $\sqrt{\mathbf{20}}$ **m**<br>$\theta$ = $\tan^{-1}(4/2)$ | **4.47 m**<br>**63.4°** *north* of *east*. |

## SECTION E THE 2026-XJ THREAD

DAY 6 OF 22 · NEW TOOL: VECTOR SUMS

XJ's motion must be tracked in more than one direction. Adding signed components keeps sideways motion in the forecast instead of treating every displacement as motion toward Earth.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Add the directions before the lengths

Vector components, resultants, and direction angles

Two trips do not combine the same way just because their lengths match. Walking four meters right and four meters left brings you back; walking both legs right does not. Direction must stay attached to displacement. Today we will use components to add vectors without losing that information.

Choose the positive horizontal axis toward east and the positive vertical axis toward north. A vector written as **(6, −4) m** means six meters in the positive horizontal direction and four meters in the negative vertical direction. These are signed components, not the vector's total length. Once the axes are chosen, every vector in the problem must use that same frame.

Keep the horizontal and vertical sums in **Section B** separate. Our first displacement is $\boldsymbol{A} = (6, -4)\ \mathbf{m}$, and the second is $\boldsymbol{B} = (-4, 8)\ \mathbf{m}$. Add matching components: $\boldsymbol{R}_x = \boldsymbol{A}_x + \boldsymbol{B}_x = 6\ \mathbf{m} - 4\ \mathbf{m} = 2\ \mathbf{m}$ and $\boldsymbol{R}_y = \boldsymbol{A}_y + \boldsymbol{B}_y = -4\ \mathbf{m} + 8\ \mathbf{m} = 4\ \mathbf{m}$. The resultant is therefore $\boldsymbol{R} = (2, 4)\ \mathbf{m}$. It describes the overall displacement from the initial point to the final point.

Trace the tip-to-tail arrows in **Section C**, then trace the dashed resultant. The second arrow starts where the first ends. A single resultant arrow connects the original starting point to the final endpoint. Moving an arrow to build this diagram does not change its length or direction. Reversing the order of the two displacements changes the intermediate point, but their final vector sum stays the same.

To find the resultant's magnitude, use the right triangle formed by its perpendicular components: $\boldsymbol{R} = \sqrt{(\boldsymbol{R}_x^2 + \boldsymbol{R}_y^2)}$. Here $\boldsymbol{R} = \sqrt{[(2\ \mathbf{m})^2 + (4\ \mathbf{m})^2]} = \sqrt{20}\ \mathbf{m} \approx 4.47\ \mathbf{m}$. Magnitude is nonnegative. Adding the individual vector magnitudes instead would describe the total length of the two-leg route, not the size of the resultant displacement.

The direction needs an angle and a reference axis. Both resultant components are positive, so the arrow lies in the upper-right quadrant. From the positive horizontal axis, $\tan\boldsymbol{\theta} = \boldsymbol{R}_y \div \boldsymbol{R}_x = 4 \div 2 = 2$, giving $\boldsymbol{\theta} \approx 63.4°$. That is 63.4° toward north from east. Saying only “63.4 degrees” would leave the direction incomplete.

I want you to check the quadrant before trusting an inverse-tangent answer. The same ratio can occur when both components are negative, even though that arrow points the opposite way. A sketch and signed components resolve the ambiguity. The resultant here should lean more upward than rightward because its vertical component is larger.

For our fictional asteroid, 2026-XJ, component addition lets us combine changes in motion while preserving direction. A large sideways component can matter as much as a component toward Earth. The calculation and the arrow diagram should identify the same final direction, not just produce a plausible-looking number.

# Acceleration, signs and turning around

**LEARNING TARGET · R4** I can find acceleration from changes in velocity.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Average $a$** | Change in velocity divided by elapsed time. |
| **Speeding up** | $v$ and $a$ have the same direction in 1D. |
| **Slowing down** | $v$ and $a$ oppose each other before the turn. |
| **Turn** | $v = 0$ for an instant; $a$ can be nonzero. |

## SECTION B KEY EQUATIONS + UNITS

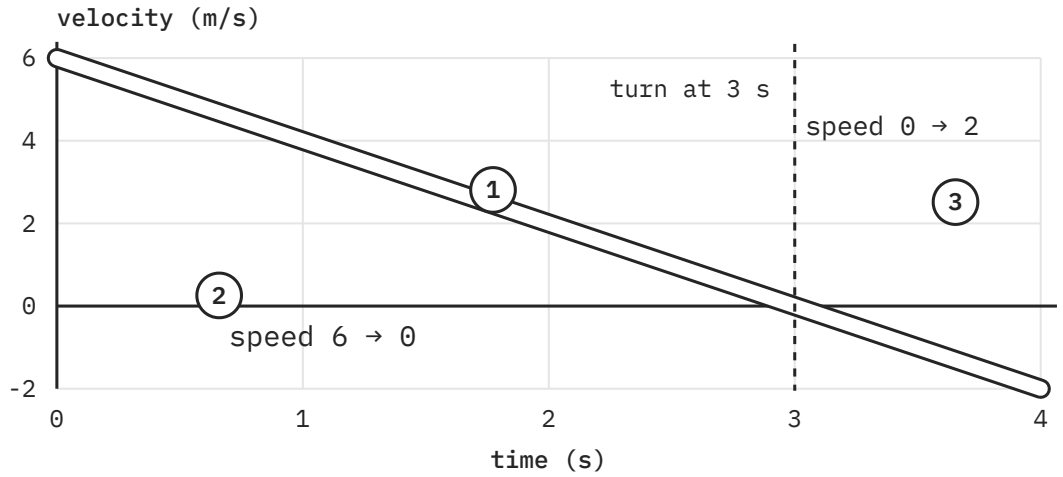
$$\bar{a} = \Delta v \div \Delta t$$

$$v_f = v_i + at$$

| | |
|---|---|
| $a$ | m/s² |
| $v$ rule | constant $a$ only |

## SECTION C COLOR IT · SLOW, THEN SPEED UP

*Right is +; constant a.*




☐ **1 · PURPLE** Signed velocity

☐ **2 · RED** Before turn: slowing

☐ **3 · GREEN** After turn: speeding up

## SECTION D WORKED EXAMPLE

A cart's velocity changes uniformly from **+6 m/s** to **−2 m/s** in **4 s**. **Find its acceleration and the time it reverses.**

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $v_i = +6$ m/s<br>$v_f = -2$ m/s<br>$\Delta t = 4$ s | $a = \Delta v/\Delta t$.<br>Turn: $0 = v_i + at$. | $a = (-2 - 6)/4$<br>$0 = 6 - 2t$<br>$t = 3$ s | $a = -2$ m/s²<br>Slows for **3 s**, then speeds up to the left. |

## SECTION E THE 2026-XJ THREAD

DAY 7 OF 22 · NEW TOOL: SIGNED CHANGE

A change in XJ's velocity requires an acceleration term in the forecast. Its sign alone does not show whether the asteroid is speeding up; compare the directions of velocity and acceleration.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Negative acceleration does not always mean slowing

Velocity changes, graph slopes, and a turning point

Imagine a cart moving right while a steady influence changes its velocity toward the left. At first the cart slows down. Later it may stop and begin moving left faster and faster. The acceleration can remain the same through that entire story. I want us to describe what changes without letting a negative sign stand in for an explanation.

Acceleration measures the rate of change of velocity: $a_{avg} = \Delta v \div \Delta t$. Its units are meters per second each second, written m/s². A negative acceleration means the velocity changes in the negative direction on our chosen axis. Whether speed increases or decreases depends on the velocity the object already has.

Take right as positive. In **Section D**, velocity changes from **+6 m/s to −2 m/s in 4 s**. The change is $\Delta v = -2\ \text{m/s} - (+6\ \text{m/s}) = -8\ \text{m/s}$. Therefore $a_{avg} = -8\ \text{m/s} \div 4\ \text{s} = -2\ \text{m/s}^2$. We are told the acceleration is constant, so this average also describes the acceleration at each moment in the interval.

Constant acceleration gives $v_f = v_i + at$. Each second, the cart's velocity decreases by 2 m/s. Starting at +6 m/s, it reaches +4, then +2, then 0, then −2 m/s at successive whole seconds. On the velocity-time graph, these values lie on a straight line. Its slope is negative because velocity decreases as time increases.

Find where the line in **Section C** crosses the time axis. That is the turning point. Setting $0\ \text{m/s} = 6\ \text{m/s} + (-2\ \text{m/s}^2)t$ gives $t = 3\ \text{s}$. During those first three seconds, velocity is positive and acceleration is negative, so speed decreases. During the next second, both are negative, so the cart speeds up in the negative direction. The red and green regions on the notes page mark that change in speed behavior.

At the exact turning instant, the cart's velocity is zero, but its acceleration is still −2 m/s². An object can be momentarily stopped while its velocity is changing. If acceleration also vanished and stayed zero at that moment, it would remain stopped. That is a different physical situation, and we should not assume it just because the velocity passes through zero.

This gives us a useful one-dimensional check. When nonzero velocity and acceleration have the same sign, speed increases. When they have opposite signs, speed decreases. The graph's slope tells us acceleration; the graph's height tells us velocity. A downward-sloping line may describe slowing on one part of the graph and speeding up on another.

For our fictional asteroid, 2026-XJ, gravity can change velocity rather than simply maintain it. The direction of that change matters. Before deciding whether the asteroid is speeding up, slowing down, or changing its path, we need both its existing velocity and its acceleration. A signed answer is useful only when we can explain the motion it describes.

# Choose an equation for steady acceleration

**LEARNING TARGET · R5** I can choose an equation for constant acceleration.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Steady $a$** | Equal velocity changes in equal times. |
| **Knowns** | List supplied values before choosing. |
| **Displacement** | Signed change in position, not distance. |
| **Cross-check** | Test with units, a graph or another equation. |

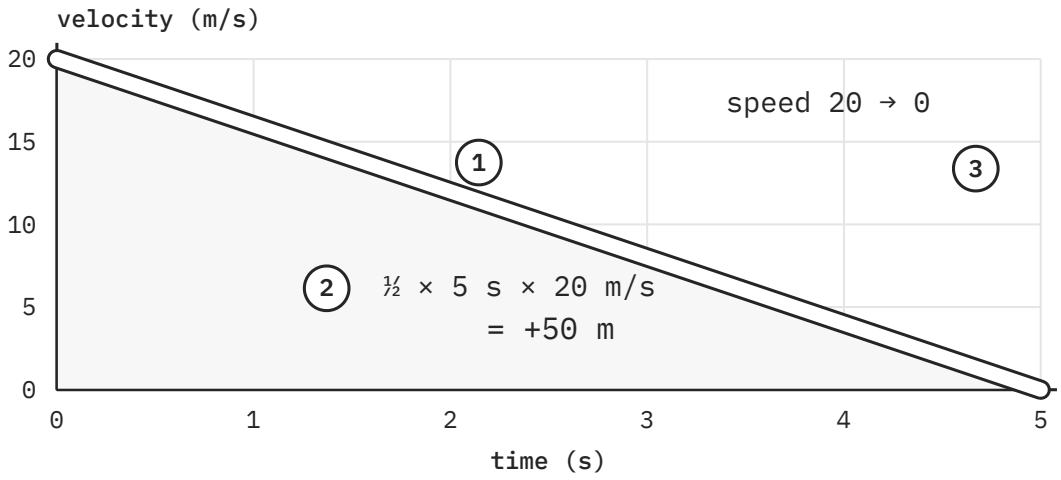
## SECTION B KEY EQUATIONS + UNITS

| Equation | Omits |
|---|---|
| $v_f = v_i + at$ | displacement |
| $\Delta x = v_i t + \frac{1}{2}at^2$ | final velocity |
| $v_f^2 = v_i^2 + 2a\Delta x$ | time |
| $\Delta x = \frac{1}{2}(v_i + v_f)t$ | acceleration |

Use for motion along one axis with constant acceleration.

## SECTION C COLOR IT · CHECK WITH AREA

*Straight line: steady a.*




☐ **1 · PURPLE** Velocity line

☐ **2 · BLUE** Displacement area

☐ **3 · RED** Decreasing speed

## SECTION D WORKED EXAMPLE

A car moving at +**20 m/s** brakes at **−4 m/s²** until it stops. **Find its stopping distance, then check the result.**

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $v_i$ = +**20 m/s**<br>$v_f$ = **0 m/s**<br>$a$ = **−4 m/s²** | No time given:<br>$v_f^2 = v_i^2 + 2a\Delta x$. | $0 = 400 - 8\Delta x$<br>$t = (0 - 20)/(-4)$<br>area = ½(20)(5) | $\Delta x$ = **+50 m**<br>**5 s** to stop; area also gives **50 m**. |

## SECTION E THE 2026-XJ THREAD

DAY 8 OF 22 · NEW TOOL: EQUATION CHOICE

For XJ, choose a relationship that fits both the known values and the motion being assumed. A graph provides a second check before a forecast is reported.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Choose the equation that fits the information

Constant acceleration, stopping distance, and independent checks

When a problem gives several numbers, it is tempting to search for an equation and start substituting. I want you to pause first. Name the starting and ending conditions, choose a positive direction, and identify the quantity you need. An equation is a statement about a model. Choosing one should follow from the information and assumptions in the situation.

Our car begins moving in the positive direction at **20 m/s** and brakes with constant acceleration **–4 m/s²** until it stops. The final velocity is **0 m/s**. We need stopping displacement, and time has not been given. Find the row marked “time” in the Omits column of **Section B**. The relation $v_f^2 = v_i^2 + 2a\Delta x$ connects exactly those quantities without requiring time.

Substitute the signed values: $(0\ \text{m/s})^2 = (20\ \text{m/s})^2 + 2(-4\ \text{m/s}^2)\Delta x$. Rearranging gives $\Delta x = [(0\ \text{m/s})^2 - (20\ \text{m/s})^2] \div [2(-4\ \text{m/s}^2)] = 50\ \text{m}$. Both numerator and denominator are negative, so displacement is positive. The units reduce to meters. Those signs tell a consistent story: the car moves forward while its forward velocity decreases.

Now find the time using $v_f = v_i + at$. Rearranging gives $t = (0 - 20)\ \text{m/s} \div (-4\ \text{m/s}^2) = 5\ \text{s}$. Negative acceleration does not require negative time or displacement. It describes the change in velocity. We stop the model at five seconds because the question asks for the first stop, not continued motion in reverse.

Trace the triangle in **Section C** to check the stopping distance from **Section D**. Constant acceleration makes velocity fall along a straight line from 20 m/s to zero. The area between that line and the time axis is a triangle: $\Delta x = \frac{1}{2}(5\ \text{s})(20\ \text{m/s}) = 50\ \text{m}$. The entire area is above zero, so displacement is positive. Because the car never reverses during this interval, its distance traveled is also 50 m.

The average of the endpoint velocities works here because acceleration is constant: $v_{avg} = (20 + 0)\ \text{m/s} \div 2 = 10\ \text{m/s}$. Multiplying that average by five seconds gives the same displacement. Equivalently, $\Delta x = \frac{1}{2}(v_i + v_f)t$. If initial velocity, acceleration, and time were given instead, $\Delta x = v_i t + \frac{1}{2}at^2$ would be convenient.

These equations describe the same constant-acceleration model from different sets of information. They are not permission to assume constant acceleration whenever numbers are available. If braking changes during the stop, we would need more information, a graph, or a model divided into appropriate intervals.

For our fictional asteroid, 2026-XJ, we must make the same decision before predicting a new position. Can velocity reasonably stay constant, or must its change be included? I want the chosen equation, signed calculation, units, and graph to support one physical story, with the model's limits stated alongside it.

# Balanced forces can mean steady motion

**LEARNING TARGET · R6** I can explain steady motion with Newton's first law.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Inertial frame** | An unforced object keeps its velocity. |
| **Net force** | Vector sum of external forces on a system. |
| **Inertia** | More mass resists velocity change more. |
| **Equilibrium** | Net force zero; $v$ constant, not always zero. |

## SECTION B KEY EQUATIONS + UNITS

$$\Sigma F = 0 \Rightarrow a = 0$$

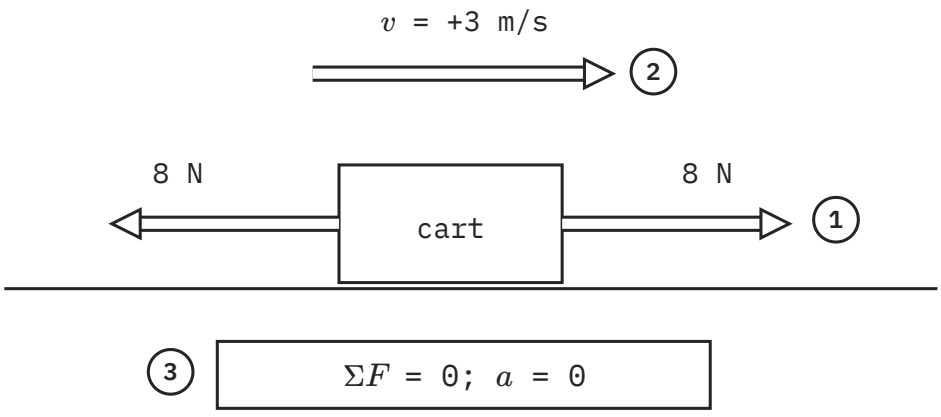
$$x_f = x_0 + v\Delta t$$

$\Sigma$ sum on one object

$v$ stays constant

## SECTION C COLOR IT · BALANCED AND MOVING

*Right is +.*




☐ **1 · ORANGE** Opposing forces

☐ **2 · PURPLE** Rightward velocity

☐ **3 · YELLOW** Net force zero

## SECTION D WORKED EXAMPLE

A cart is at **−2 m**, moving at **+3 m/s**. Its external forces balance for the next **4 s**. **Find its final position.**

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $x_0$ = **−2 m**<br>$v_i$ = **+3 m/s**<br>$\Sigma F$ = 0; $t$ = **4 s** | First law: constant $v$.<br>$x_f = x_0 + vt$. | $a$ = **0 m/s²**<br>$x_f$ = −2 + 3(4) | **+10 m**<br>Net force zero; motion continues. |

## SECTION E THE 2026-XJ THREAD

DAY 9 OF 22 · NEW TOOL: INERTIA

XJ needs no forward force simply to keep moving. Gravity can still change its velocity, so treating a short interval as steady motion is an approximation to be tested.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Balanced forces can go with steady motion

Newton's first law, inertia, and equilibrium

A book sliding across a table eventually stops. That everyday observation can make it seem as if motion needs a continuing forward force. But the book is also interacting with the table and the air. To understand why its motion changes, we need to identify those interactions rather than assume stopping is what all moving objects naturally do.

Newton's first law says that an object remains at rest or moves with constant velocity when the net external force on it is zero, viewed from an inertial frame. An inertial frame is a nonaccelerating reference frame in which this law holds. For our classroom examples, we treat the room as such a frame. Constant velocity means both speed and direction stay unchanged.

The net force is the vector sum of all forces acting on the chosen object. Individual forces can be present while their sum is zero. Compare the horizontal arrows on the cart in **Section C**. Here **8 N right** balances **8 N left**. The upward and downward forces balance too. Thus $\Sigma \boldsymbol{F} = \mathbf{0}$ and $\boldsymbol{a} = \mathbf{0}$. The symbol Σ means add all the relevant forces, including their directions.

Zero acceleration tells us velocity does not change. It does not tell us that velocity is zero. Suppose the cart is already moving at **+3 m/s**. With balanced forces, it continues at +3 m/s. If that same balanced-force situation began with the cart at rest, it would remain at rest. We need the initial velocity to distinguish those two cases.

The example in **Section D** starts the cart at **−2 m** and keeps the forces balanced for **4 s**. Its displacement is $\Delta \boldsymbol{x} = (3\ \mathrm{m/s})(4\ \mathrm{s}) = +12\ \mathrm{m}$. Using $\boldsymbol{x_f} = \boldsymbol{x_0} + \boldsymbol{vt}$, we obtain $\boldsymbol{x_f} = -2\ \mathrm{m} + 12\ \mathrm{m} = 10\ \mathrm{m}$. The cart changes position even though its acceleration and net force are zero.

We call the resistance to a change in velocity inertia. It is a property of matter, not an extra force that should be drawn pushing the cart forward. Mass measures inertia: a larger mass requires a larger net force to produce the same acceleration. In the balanced situation, however, neither a small nor a large mass accelerates just because it is moving.

The vocabulary in **Section A** calls this balance equilibrium. It can describe rest or steady straight-line motion. The yellow code on the notes page marks that balance; purple identifies velocity. Keeping those ideas separate also explains the sliding book. Friction supplies a force opposing its sliding motion, so the horizontal forces do not balance and its velocity changes. A continuing push can maintain steady motion when it balances that friction.

For our fictional asteroid, 2026-XJ, motion through space does not require a forward force to keep it going. Gravity can still provide a net force and change its velocity, including its direction. I want us to ask what forces change the motion, then use the existing velocity to explain what the object actually does.

# Name each force and its source

**LEARNING TARGET · K2** I can name each force, its source and its receiver.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Weight** $W$ | Earth's downward pull: $mg$. |
| **Normal** $N$ | Surface push, perpendicular to contact. |
| **Tension** $T$ | Rope's pull along itself, away from the object. |
| **Receiver** | The object a force acts on; also name its source. |

## SECTION B KEY EQUATIONS + UNITS

$$W = mg$$

$$N + T - W = 0$$

| | |
|---|---|
| $g$ | **9.8 N/kg** |
| balance | vertical $a = 0$ |

## SECTION C COLOR IT · ONE CRATE, THREE FORCES

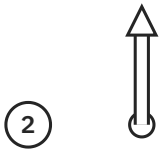
*A rope helps a table.*

rope on crate




$T$ = 10 N

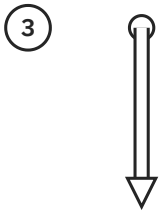
table on crate




$N$ = 19.4 N

Earth on crate

$W$ = 29.4 N




Three forces; one receiver. The crate stays at rest.

- ☐ **1 · ORANGE** Rope's tension $T$
- ☐ **2 · ORANGE** Table's normal $N$
- ☐ **3 · ORANGE** Earth's weight $W$

## SECTION D WORKED EXAMPLE

A **3 kg** crate rests on a table. A vertical rope pulls up with **10 N**. **Find the table's normal force.** Use $g$ = **9.8 N/kg**.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $m$ = **3 kg**<br>$T$ = **10 N** up<br>$a_y$ = 0 | $W = mg$.<br>$N + T - W = 0$. | $W$ = 3(9.8) = **29.4 N**<br>$N$ = 29.4 − 10 | $N$ = **19.4 N**<br>The rope shares the support. |

## SECTION E THE 2026-XJ THREAD

DAY 10 OF 22 · NEW TOOL: FORCE SOURCES

A forecast of XJ's changing motion needs real interactions, not a made-up motion force. Name the attracting body for each gravitational force before adding the forces.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Every force has a source and a receiver

Weight, normal force, tension, and supported objects

Before we add forces, we need to know which forces belong in the calculation. A useful habit is to name both the object receiving a force and the object exerting it. “The rope pulls on the crate” identifies an interaction. “The crate is moving” describes its motion, but does not identify a new force. I want each arrow to have a physical source.

Weight is the gravitational force Earth exerts on an object. Near Earth's surface, its magnitude is $W = mg$. Mass is measured in kilograms; weight is measured in newtons. We use $g = 9.8\ \text{N/kg}$, meaning each kilogram has 9.8 N of gravitational force acting on it in this model. The same value can be expressed as 9.8 m/s² when describing free-fall acceleration.

The normal force comes from contact with a surface and acts perpendicular to that surface. Tension comes from a taut rope and pulls along the rope, away from the object. Neither force is automatically equal to weight. Its magnitude depends on the other forces and on the object's acceleration. A surface and a rope can share the job of supporting an object.

Follow the given values in **Section D** for the **3 kg crate** resting on a level table while a vertical rope pulls upward with **10 N**. Earth's downward pull is $W = (3\ \text{kg})(9.8\ \text{N/kg}) = 29.4\ \text{N}$. The table pushes upward with a normal force whose magnitude we have not yet found. There are three forces on the crate: weight, tension, and the normal force.

The crate stays at rest, so its vertical acceleration is zero. Choose upward as positive. The force equation is $N + T - W = 0$. Substituting gives $N + 10\ \text{N} - 29.4\ \text{N} = 0$, so $N = 19.4\ \text{N}$. Here italic N represents the normal-force magnitude; upright N after a number is the unit newton. The rope supplies part of the upward support, leaving the table to supply the rest.

Trace each of the three arrows in **Section C** back to its named source. The crate receives all three forces. Orange identifies force. If the arrows use one consistent scale, the two upward lengths together should match the downward length. The normal arrow should be longer than the tension arrow, but shorter than the weight arrow. That picture is a useful check on the numbers and the balance equation.

If the rope were absent and the crate remained at rest, the normal force would equal the weight. That equality follows from the particular situation; it is not the definition of a normal force. An inclined surface or an additional vertical force changes the calculation. Likewise, mass remains the same when support changes, even though a scale's reading can change.

For our fictional asteroid, 2026-XJ, we will name the source of every proposed force before using it. Gravity can act without contact. A rope, supporting surface, or engine force belongs only when that interaction exists. A clear list of sources and receivers prevents us from inventing forces simply to match the motion we expect.

# Change mass while net force stays fixed

**LEARNING TARGET · S3** I can test how mass affects acceleration at fixed force.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Mass** $M$ | All moving mass, including any hanger. |
| **Control** | A value held fixed for a fair comparison. |
| **Inverse** | Double $M$ halves $a$ at fixed net force. |
| **Slope** | Plot $a$ against $1/M$ to get an ideal line. |

## SECTION B KEY EQUATIONS + UNITS

$$a = F_{\text{net}} \div M$$

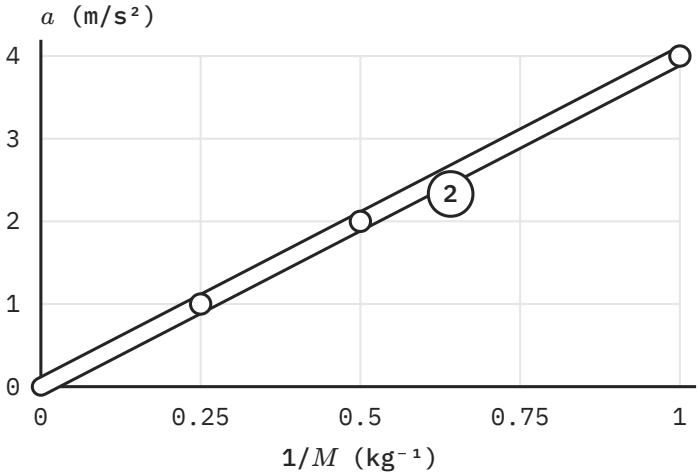
$$M \times a = F_{\text{net}}$$

$1/M$ inverse mass, $kg^{-1}$

slope units: N

## SECTION C COLOR IT · CHANGE ONLY MASS

*Ideal data; net force **4 N**.*




| | $M$ (kg) | $a$ (m/s²) |
|---|---|---|
| 1 | 1 | 4 |
| | 2 | 2 |
| | 4 | 1 |

3 $F_{\text{net}}$ = 4 N

☐ **1 · PENCIL** Mass and data points

☐ **2 · GREEN** Speeding up: moving +

☐ **3 · ORANGE** Fixed net force

## SECTION D WORKED EXAMPLE

Under the same net force, a **1 kg** system accelerates at **4 m/s²** and a **2 kg** system at **2 m/s²**. **Predict $a$ for a 3 kg system.**

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $M_1$ = **1 kg**; $a_1$ = **4 m/s²**<br>$M_2$ = **2 kg**; $a_2$ = **2 m/s²**<br>same net force | Check $M \times a$.<br>Then $a = F_{\text{net}}/M$. | 1(4) = 2(2) = **4 N**<br>$a_3$ = **4 N** ÷ **3 kg** | **1.33 m/s²**<br>Assumes net force stays **4 N**. |

## SECTION E THE 2026-XJ THREAD

DAY 11 OF 22 · NEW TOOL: MASS COMPARISON

For the same force, a more massive XJ changes velocity less quickly. The comparison needs the same net force; a larger gravitational force can change that result.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# The same force can produce different accelerations

Total moving mass, inverse relationships, and experimental controls

Suppose I use the same net force to accelerate two carts, but one carries more mass. The heavier system changes velocity more slowly. That comparison is useful only if the net force really stays the same. Today we are looking at how acceleration depends on mass while holding the force constant, and how a graph can make that relationship visible.

Newton's second law connects the quantities: $\boldsymbol{F_{net} = Ma}$, or $\boldsymbol{a = F_{net} \div M}$. Here the capital M means the total mass of the system we are accelerating. In a cart-and-hanger arrangement, the hanger's mass belongs in that total if it moves as part of our chosen system. Counting only the cart would make our model incomplete.

Compare the mass and acceleration columns in **Section C**. They show **1 kg accelerating at 4 m/s²** and **2 kg accelerating at 2 m/s²**. Each trial gives the same product: **(1 kg)(4 m/s²) = (2 kg)(2 m/s²) = 4 N**. That common product is the net force. Doubling mass halves acceleration when the net force is unchanged.

Use the mass in **Section D** for the next prediction. For a **3 kg** system under that same net force, $\boldsymbol{a = 4\ N \div 3\ kg \approx 1.33\ m/s^2}$. For 4 kg, the acceleration would be 1 m/s². These values form an inverse relationship. Equal additions of mass do not produce equal decreases in acceleration. The graph of acceleration against mass therefore bends instead of forming a straight line.

We can reveal the pattern by plotting acceleration against reciprocal mass, meaning one divided by mass. The equation becomes $\boldsymbol{a = F_{net}(1/M)}$. This graph is a straight line through the origin for the ideal model. Its slope is the fixed net force. The horizontal units are inverse kilograms, and dividing m/s² by inverse kilograms gives kg·m/s², or newtons.

That graph is more than a way to make the data look tidy. It tests a specific prediction: acceleration should be proportional to reciprocal mass under constant net force. If the data do not follow the predicted line within measurement uncertainty, we should examine the setup and assumptions. Friction, a changing pull, or an incorrectly counted moving mass can change the relationship we actually measured.

I want us to distinguish a controlled variable from something we merely hope stayed constant. Adding mass may also change friction. Moving mass between a cart and a hanger may preserve total mass while changing the driving force. The force diagram and system boundary help us decide whether two trials truly test the same relationship. Orange represents the applied forces; the measured change in velocity supplies acceleration.

For our fictional asteroid, 2026-XJ, a fixed proposed push would produce less acceleration for a larger mass. But this does not mean more massive objects automatically fall more slowly under gravity. Gravitational force also depends on mass. We must check what is held constant before carrying a relationship from one physical situation into another.

# Change force while total mass stays fixed

**LEARNING TARGET · S4** I can test $F = ma$ while keeping total mass fixed.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Driving force** | The pull before resistance is subtracted. |
| **Net force** | Driving force minus opposing resistance. |
| **Slope** | On an $a$-versus-net-force graph: $1/M$. |
| **Offset** | Fixed friction shifts the driving-force line down. |

## SECTION B KEY EQUATIONS + UNITS

$$a = F_{\text{net}}/M$$

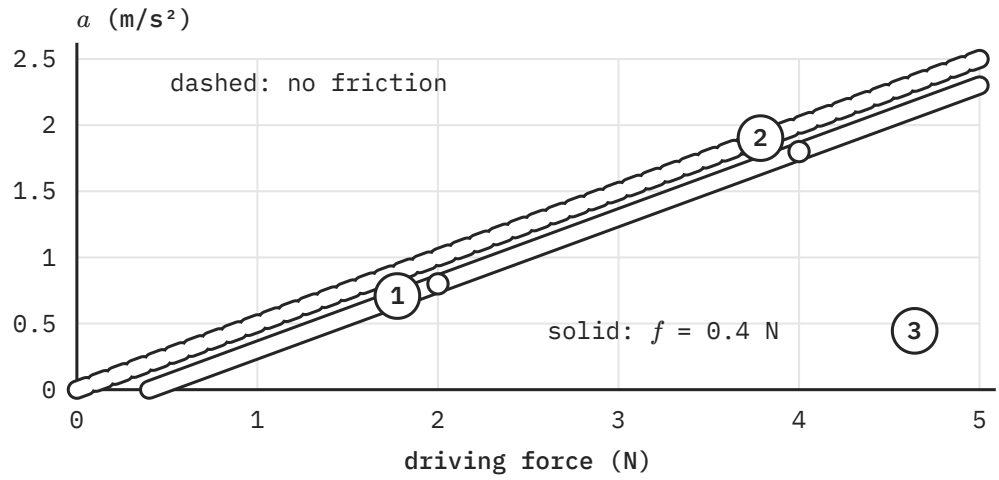
$$F_{\text{net}} = F_{\text{drive}} - f$$

$M$ total mass, kg

N/kg equals m/s²

## SECTION C COLOR IT · DRIVING VS. NET FORCE

*Moving +; $M$ = **2 kg**.*




☐ **1 · PENCIL** Axes and data

☐ **2 · GREEN** Speeding up here

☐ **3 · ORANGE** Force labels

## SECTION D WORKED EXAMPLE

A **2 kg** system slides with **0.4 N** of opposing friction. **Compare acceleration with driving forces of 2 N and 4 N.**

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $M$ = **2 kg**<br>$f$ = **0.4 N**<br>$F_{\text{drive}}$ = **2 N**, then **4 N** | Subtract friction.<br>Then $a = F_{\text{net}}/M$. | $a_1 = (2 - 0.4)/2 = 0.8$<br>$a_2 = (4 - 0.4)/2 = 1.8$ | **0.8; 1.8 m/s²**<br>Driving force doubles; $a$ does not. |

## SECTION E THE 2026-XJ THREAD

DAY 12 OF 22 · NEW TOOL: NET-FORCE TEST

A proposed push on XJ must be added to every other external force. Only the net force belongs in the calculation of acceleration.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# An applied pull is not always the net force

Fixed mass, friction, and force-acceleration graphs

If we double a pulling force, will acceleration double? It can, but only when the net force doubles and the mass stays fixed. The force we apply may have to overcome another force acting in the opposite direction. Today I want us to keep the applied force and the sum of all forces separate from the start.

Our model is a **2 kg** object sliding in the positive horizontal direction. Friction acts backward with a constant magnitude of **0.4 N**. The vertical forces balance. Horizontally, $F_{net} = F_{drive} - f$, and Newton's second law gives $a = F_{net} \div M$. The signs describe the forces along our chosen axis.

Start with the first driving force in **Section D**. With a **2 N** driving force, the net force is $F_{net} = 2.0\ \text{N} - 0.4\ \text{N} = 1.6\ \text{N}$. Therefore $a = 1.6\ \text{N} \div 2\ \text{kg} = 0.8\ \text{m/s}^2$. The backward friction force reduces the velocity change produced each second, but the net force is still forward, so acceleration is positive.

Now increase the drive to **4 N** while keeping the mass and sliding-friction model unchanged. The net force becomes $F_{net} = 4.0\ \text{N} - 0.4\ \text{N} = 3.6\ \text{N}$, and $a = 3.6\ \text{N} \div 2\ \text{kg} = 1.8\ \text{m/s}^2$. The drive doubled, but acceleration changed from 0.8 to 1.8 m/s². It did not merely double to 1.6 m/s² because the net force did not merely double.

Plotting acceleration against net force produces a straight line through the origin for fixed mass. Its slope is $1/M = 1/(2\ \text{kg}) = 0.5\ \text{kg}^{-1}$. Each extra newton of net force adds 0.5 m/s² of acceleration. The graph gives another way to estimate mass: a larger mass produces a shallower acceleration-versus-net-force slope.

Compare the solid and dashed lines in **Section C**. Plotting against the driving force still gives a straight line in this sliding model, but it is shifted by friction. The equation is $a = (F_{drive} - f)/M$. Its mathematical acceleration intercept is negative because friction is being subtracted. We should not extend that line blindly to an object resting on the table: static friction behaves differently, and the direction of sliding matters too.

The experimental question is whether mass and opposing forces stayed consistent across trials. If we change the hanger, load, or surface, we may change more than the intended driving force. In the notes diagram, compare the orange force arrows before interpreting the acceleration values. The difference between the forward and backward arrows is what enters Newton's second law.

For our fictional asteroid, 2026-XJ, a proposed engine or push would be one interaction in a larger force picture. We would add it to gravity and any other relevant forces to find the net change. I want our prediction to follow from that complete sum, with the mass and model assumptions stated clearly.

# Net force first, then interpret the motion

**LEARNING TARGET · R7** I can solve with $F = ma$ — always finding the NET force first.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Force sum** | Add signed forces along the same axis. |
| **Acceleration** | Velocity change from net force per mass. |
| **Velocity** | Current motion; it can oppose $a$. |
| **Speed trend** | Opposite $v$ and $a$: slows before reversal. |

## SECTION B KEY EQUATIONS + UNITS

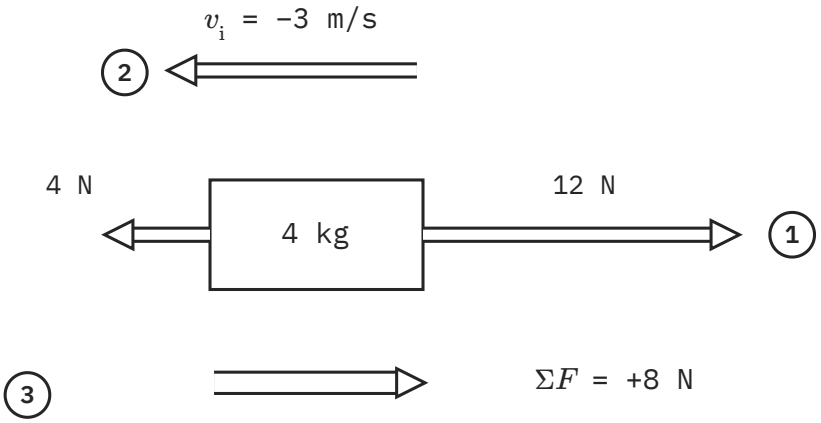
$$\Sigma F_x = ma_x$$

$$v_f = v_i + a_x t$$

| | |
|---|---|
| $\Sigma F_x$ | $F_{right} - F_{left}$ |
| + | right |

## SECTION C COLOR IT · FORCE OPPOSES MOTION

*The cart moves left.*




after 1 s: $v$ = −1 m/s; speed is smaller

- ☐ **1 · ORANGE** — Forces and separate net
- ☐ **2 · PURPLE** — Leftward velocity
- ☐ **3 · RED** — Slows for the first **1 s**

## SECTION D WORKED EXAMPLE

A **4 kg** cart has forces of **12 N** right and **4 N** left. Its initial velocity is **−3 m/s**. **Find $a$ and $v$ after 1 s.**

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $m$ = **4 kg**<br>$F_{right}$ = **12 N**; $F_{left}$ = **4 N**<br>$v_i$ = **−3 m/s** | Sum forces first.<br>$\Sigma F_x = ma_x$.<br>$v = v_i + a_x t$. | $\Sigma F_x$ = 12 − 4 = **+8 N**<br>$a_x$ = 8/4 = **+2 m/s²**<br>$v$ = −3 + 2(1) | $v$ = **−1 m/s**<br>Still leftward; speed is now **1 m/s**. |

## SECTION E THE 2026-XJ THREAD

DAY 13 OF 22 · NEW TOOL: FORCE TO MOTION

The direction of XJ's net force tells us its acceleration direction. Its existing velocity must also be known to say whether it speeds up, slows down or turns.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# The force tells us how motion changes

Net force, acceleration, and a cart moving the other way

A force arrow pointing right does not guarantee that the object is moving right. It tells us about an interaction. The net force determines acceleration, and acceleration tells us how velocity changes. To describe the motion itself, we also need the velocity the object already has. Today we will follow that chain carefully.

Choose right as positive. Our **4 kg cart** has a **12 N force to the right** and a **4 N force to the left**. Both forces act on the cart, so they belong in the same horizontal sum: $\Sigma F_x = 12\ \text{N} - 4\ \text{N} = +8\ \text{N}$. Using only the larger force would ignore a real interaction. Adding both magnitudes without signs would ignore their opposite directions.

Use the force sum in **Section B** with right as positive: $\Sigma F_x = ma_x$. Thus $a_x = +8\ \text{N} \div 4\ \text{kg} = +2\ \text{m/s}^2$. The acceleration points right, matching the net force. The vertical forces are balanced, so they produce no vertical acceleration and do not add to the horizontal force sum.

Now include the starting velocity, **−3 m/s**. The cart initially moves left. With constant forces and mass, acceleration remains constant, so after **1 s** we use $v_f = v_i + a_x t$. Substitution gives $v_f = -3\ \text{m/s} + (2\ \text{m/s}^2)(1\ \text{s}) = -1\ \text{m/s}$. The cart still moves left, but its speed has fallen from 3 m/s to 1 m/s.

This is what positive acceleration means here: the signed velocity becomes more positive by 2 m/s each second. It does not instantly become positive. From −3 to −1 is an increase in velocity but a decrease in speed. I want you to say both parts when interpreting the answer, because the sign and the size describe different features of the motion.

If the same forces continue, the cart reaches zero velocity after $t = (0 - (-3))\ \text{m/s} \div (2\ \text{m/s}^2) = 1.5\ \text{s}$. It then moves right and speeds up. There is no need for the net force to reverse at the turning point. The same rightward acceleration first slows the leftward motion and then builds rightward motion.

In **Section C**, compare the cart's force and velocity arrows. The orange arrows identify forces and the purple arrow identifies initial velocity. The red label applies to the first second because speed decreases during that interval. The net-force arrow is a summary of the force sum, shown separately; it is not an additional force to add again. Keeping the picture and calculation connected helps prevent double-counting.

For our fictional asteroid, 2026-XJ, a force toward Earth tells us the direction of its acceleration due to that interaction. Its existing velocity might have components toward, away from, or across that direction. A useful forecast combines the net force with that initial motion and an elapsed time, rather than assuming the asteroid immediately travels along the force arrow.

# Equal forces, different accelerations

**LEARNING TARGET · R8** I can identify force pairs acting on different objects.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Force pair** | A-on-B and B-on-A: equal and opposite. |
| **Receivers** | Two objects; use separate force diagrams. |
| **Mass effect** | Same force size: smaller mass, larger $a$. |
| **System** | Pair forces cancel internally for combined A+B. |

## SECTION B KEY EQUATIONS + UNITS

$$F_{A\to B} = -F_{B\to A}$$

equal and opposite

$a_A$ $F_A/m_A$
$a_B$ $F_B/m_B$

## SECTION C COLOR IT · TWO RECEIVERS

*Same **6 N** force size.*

Equal force sizes on different receivers




$a_A = -6$ m/s²

$a_B = +2$ m/s²

☐ **1 · ORANGE** Force on A: left

☐ **2 · ORANGE** Force on B: right

☐ **3 · PENCIL** Objects and masses

## SECTION D WORKED EXAMPLE

Cart A has mass **1 kg**; cart B has mass **3 kg**. Their interaction pushes A left and B right with **6 N** each. **Find both accelerations.**

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $m_A$ = **1 kg**<br>$m_B$ = **3 kg**<br>right is + | $F = ma$ on each cart.<br>No other horizontal forces. | $a_A$ = −6/1 = **−6 m/s²**<br>$a_B$ = +6/3 = **+2 m/s²** | **−6; +2 m/s²**<br>A has 3 times B's acceleration size. |

## SECTION E THE 2026-XJ THREAD

DAY 14 OF 22 · NEW TOOL: INTERACTION PAIRS

Earth and fictional XJ pull on each other equally at the same instant. Their very different masses give different accelerations; the pair does not cancel in XJ's own force diagram.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Equal forces can have unequal effects

Newton's third law, separate objects, and system boundaries

When two objects interact, each exerts a force on the other. That statement applies even when one object is much larger or appears to do most of the moving. Newton's third law describes the two forces in a single interaction: they have equal magnitudes and opposite directions, and they act at the same time on different objects.

Read the object names on either side of the equation in **Section B**: $\boldsymbol{F}_{\mathbf{A\ on\ B}} = -\boldsymbol{F}_{\mathbf{B\ on\ A}}$. The labels matter as much as the minus sign. One force is received by B; the other is received by A. The force on B does not belong in A's individual force sum. I want us to identify the receivers before deciding whether two arrows can cancel in a calculation.

Our example has cart A with mass **1 kg** and cart B with mass **3 kg**. Their interaction pushes A left with **6 N** and B right with **6 N**. Take right as positive and assume no other horizontal forces. For A, $\boldsymbol{a}_{\mathbf{A}} = \mathbf{-6\ N \div 1\ kg = -6\ m/s^2}$. For B, $\boldsymbol{a}_{\mathbf{B}} = \mathbf{+6\ N \div 3\ kg = +2\ m/s^2}$.

The forces are equal in magnitude, but A's acceleration has three times B's magnitude. The smaller mass undergoes a larger velocity change each second under the same force size. That difference does not violate the third law. The third law compares the interaction forces; the second law explains how each object's mass responds to its own net force.

Find the receiver of each force in **Section C**. The diagram uses two equal-length orange arrows pointing opposite ways. A separate force diagram for A includes the leftward force on A. A separate diagram for B includes the rightward force on B. If we put both arrows on one cart merely because they form a pair, we would incorrectly predict no acceleration for that cart.

We can instead choose the combined A-and-B system. Then this interaction is internal: both members of the pair act between parts inside our chosen boundary. Their contributions cancel in the combined system's net external-force calculation. The carts can still accelerate relative to one another. With no external horizontal force, the combined system's center-of-mass velocity stays constant.

A third-law pair is not just any two opposite forces of equal size. Weight and a table's normal force on a resting book both act on the book, so they are not partners. The partner of Earth's pull on the book is the book's pull on Earth. The partner of the table's push on the book is the book's push on the table. Naming sources and receivers makes the distinction clear.

For our fictional asteroid, 2026-XJ, Earth pulls on the asteroid and the asteroid pulls on Earth with equal force magnitudes at the same instant. Their very different masses produce very different accelerations. Earth's pull belongs in the asteroid's force diagram; the asteroid's pull on Earth does not cancel it there. Our system boundary determines which forces enter each prediction.

# Draw forces on one chosen object

**LEARNING TARGET · S5** I can draw every force on one object, with arrows to scale.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Object** | The receiver of every force in the diagram. |
| **External** | A force from outside the chosen boundary. |
| **Length** | Use one scale to show each force's size. |
| **Net force** | Sum of forces, not another interaction. |

## SECTION B KEY EQUATIONS + UNITS

$$T - mg = ma_y$$

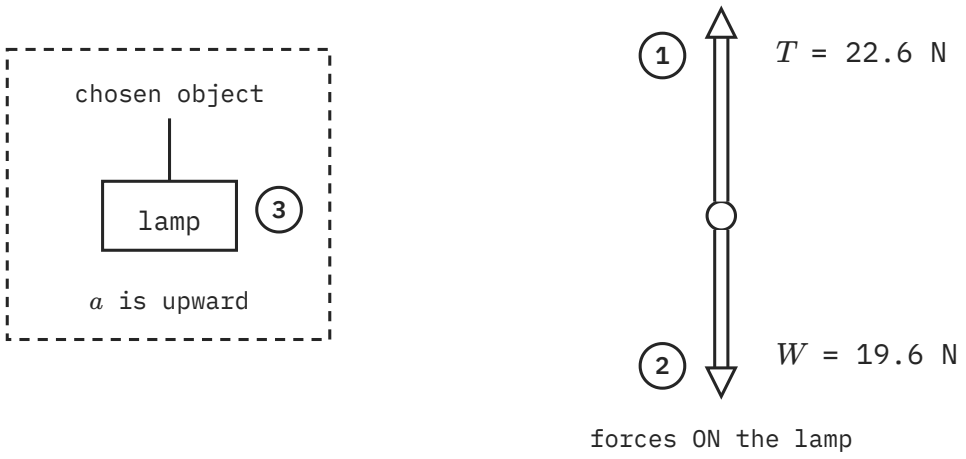
$$T = m(g + a_y)$$

\+ up

$W$ Earth on lamp

## SECTION C COLOR IT · LAMP ACCELERATES UP

*Forces on the lamp only.*




- ☐ **1 · ORANGE** Tension: longer arrow
- ☐ **2 · ORANGE** Weight: shorter arrow
- ☐ **3 · PENCIL** Lamp and boundary

## SECTION D WORKED EXAMPLE

A rope lifts a **2 kg** lamp with upward acceleration **1.5 m/s²**. **Find the tension.** Use $g$ = **9.8 m/s²**.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $m$ = **2 kg**<br>$a_y$ = **+1.5 m/s²**<br>$g$ = **9.8 m/s²** | Up is +.<br>$T - mg = ma_y$. | $W$ = 2(9.8) = **19.6 N**<br>$T$ = 19.6 + 2(1.5) | $T$ = **22.6 N**<br>$T$ and $W$ are not a third-law pair. |

## SECTION E THE 2026-XJ THREAD

DAY 15 OF 22 · NEW TOOL: OBJECT BOUNDARY

An XJ force diagram contains forces on XJ, not the forces XJ exerts on other bodies. Defining that boundary keeps the acceleration calculation consistent.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# A force diagram begins with one chosen object

External interactions, scaled arrows, and an accelerating lamp

A useful force diagram leaves out many visible details so that the interactions become easier to see. The first decision is the object or system we are studying. Every force arrow in that diagram must act on that chosen receiver. Today I want us to draw that boundary before writing Newton's second law.

Consider a **2 kg lamp** hanging from a taut vertical rope. The lamp accelerates upward at **1.5 m/s²**. We choose the lamp as our object. The rope exerts upward tension on it, and Earth exerts downward weight. Those are the two external forces in this model. The force the lamp exerts on the rope belongs on the rope's diagram, not on the lamp's.

Use the positive direction in **Section B**, upward. The lamp's weight magnitude is $W = mg = (2\ \text{kg})(9.8\ \text{m/s}^2) = 19.6\ \text{N}$. Because weight acts downward, it enters the vertical sum with a minus sign. Tension acts upward and enters with a plus sign. Newton's second law becomes $T - mg = ma_y$.

Rearranging gives $T = mg + ma_y$. Substituting, $T = 19.6\ \text{N} + (2\ \text{kg})(1.5\ \text{m/s}^2) = 22.6\ \text{N}$. The rope must supply enough upward force to balance the weight and an additional 3.0 N to produce the upward acceleration. The net force is therefore 3.0 N upward, not 22.6 N.

Compare the lengths of the two force arrows in **Section C**. Draw the upward tension arrow longer than the downward weight arrow, using the same length scale for both. Their difference represents the net force. A separately labeled net-force arrow can summarize the calculation, but it is not a third interaction to add to tension and weight. Adding it again would count the same imbalance twice.

Tension and weight are not a third-law pair. They both act on the lamp, come from different sources, and here even have different magnitudes. The tension's partner is the lamp's pull on the rope. The weight's partner is the lamp's gravitational pull on Earth. Neither partner belongs in this single-object diagram.

An upward acceleration also does not, by itself, prove that the lamp is moving upward. It could be moving downward and slowing. If it is already moving upward, it speeds up. The forces determine the change in velocity; a starting velocity is still needed to determine the motion. Any velocity arrow should be kept distinct from the orange force arrows so it cannot be mistaken for another force.

For our fictional asteroid, 2026-XJ, we will use exactly this discipline. Define the asteroid as the receiver, list the external interactions on it, and draw their directions before combining them. An uncluttered diagram is useful because it makes missing forces, incorrect receivers, and double-counted components easier to catch before they affect the forecast.

# Choose useful axes on an incline

**LEARNING TARGET · R9** I can use a force diagram to find acceleration on a ramp.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Parallel ∥** | Along the ramp; choose downhill as +. |
| **Normal axis ⊥** | Straight out of the ramp, usually not vertical. |
| **Weight parts** | $mg \sin\theta$ along; $mg \cos\theta$ into the ramp. |
| **Constraint** | No motion through the surface: $a_\perp = 0$. |

## SECTION B KEY EQUATIONS + UNITS

$$ma_\parallel = mg \sin\theta - f$$

$$N = mg \cos\theta$$

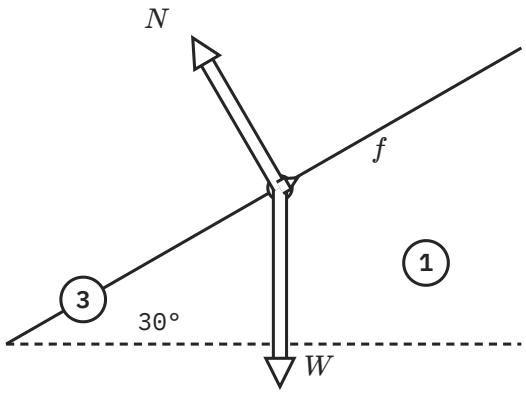
$\theta$ ramp angle

$N$ rule no other ⊥ forces

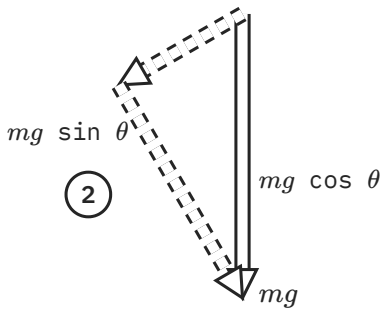
## SECTION C COLOR IT · FORCES AND COMPONENTS

*Keep the two views separate.*

actual forces




resolve $W$ separately




☐ **1 · ORANGE** Actual forces

☐ **2 · ORANGE** Weight's components

☐ **3 · PENCIL** Ramp and angle

## SECTION D WORKED EXAMPLE

A **2 kg** block slides down a **30°** ramp with **2 N** friction uphill. **Find its acceleration and normal force.** Use $g$ = **9.8 m/s²**.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $m$ = **2 kg**; $\theta$ = **30°**<br>$f$ = **2 N** uphill<br>$a_\perp$ = 0 | $mg \sin\theta - f = ma_\parallel$<br>$N = mg \cos\theta$ | $a_\parallel$ = [19.6(0.5) − 2]/2<br>$N$ = 19.6 cos **30°** | **3.9 m/s²**<br>Downhill; $N \approx$ **17.0 N**. |

## SECTION E THE 2026-XJ THREAD

DAY 16 OF 22 · NEW TOOL: CHOSEN AXES

A suitable set of axes can simplify the forces changing XJ's path. Components are calculation tools; they must not be counted again as extra forces.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Choose axes that help explain the ramp

Inclined surfaces, weight components, and normal force

A block on a ramp still has weight acting straight down, but its motion follows the slope. We can choose axes that make that motion easier to describe. Let the parallel axis point downhill and the perpendicular axis point outward from the surface. Rotating our axes does not change any physical force. It changes how we describe the force's components.

Our **2 kg block** slides down a ramp inclined **30° above horizontal**. Friction acts uphill with magnitude **2 N**. The actual forces are weight downward, normal force perpendicular away from the ramp, and friction along the ramp. With $g = 9.8\ \text{m/s}^2$, the weight magnitude is $W = mg = 19.6\ \text{N}$.

Match each component in **Section B** to an axis along or perpendicular to the ramp. Its downhill component has magnitude $W_{\text{parallel}} = mg \sin\theta$, while its component into the surface has magnitude $W_{\text{into}} = mg \cos\theta$. The ramp angle determines this component triangle. A useful check is a nearly level ramp: the downhill component should approach zero, while nearly all the weight points into the surface.

For our angle, the downhill part is $(19.6\ \text{N}) \sin 30° = 9.8\ \text{N}$. Friction opposes it, so $\Sigma F_{\text{parallel}} = 9.8\ \text{N} - 2.0\ \text{N} = 7.8\ \text{N}$. Newton's second law then gives $a_{\text{parallel}} = 7.8\ \text{N} \div 2\ \text{kg} = 3.9\ \text{m/s}^2$ downhill. Because the block is already sliding downhill, this acceleration makes it speed up.

The block does not accelerate through the solid ramp or away from it in this model. Its perpendicular acceleration is zero. Taking outward as positive gives $N - mg \cos\theta = 0$, so $N = (19.6\ \text{N}) \cos 30° \approx 17.0\ \text{N}$. This normal force is smaller than the total weight. It balances only the component pressing into the ramp.

That normal-force equation depends on the situation. It works because there are no other perpendicular forces and no perpendicular acceleration. An angled pull or a curved surface could change the calculation. I want you to begin with the force balance instead of assuming normal force always equals either weight or one memorized expression.

Compare the two views in **Section C**. The drawing separates the actual-force diagram from the weight-component triangle. This distinction prevents a common mistake: adding weight and then adding both of its components as if three separate interactions existed. The components replace weight in the calculation; they do not supplement it. Orange marks force information in both views, while the ramp and axes remain in pencil.

For our fictional asteroid, 2026-XJ, useful axes might follow a proposed path or point toward a gravitational source. Whatever axes we choose, the resulting prediction should describe the same physical motion. Components make the mathematics manageable, but the real interactions and the chosen system must remain consistent throughout the calculation.

# Static and kinetic friction

**LEARNING TARGET** I can distinguish adjustable static friction from kinetic friction.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Static** $f_s$ | Adjusts to prevent slip, up to a limit. |
| **Limit** | $f_s$,max = $\mu_s N$; $f_s$ can be smaller. |
| **Kinetic** $f_k$ | Opposes sliding; model size $\mu_k N$. |
| **Normal** $N$ | Find the perpendicular force balance first. |

## SECTION B KEY EQUATIONS + UNITS

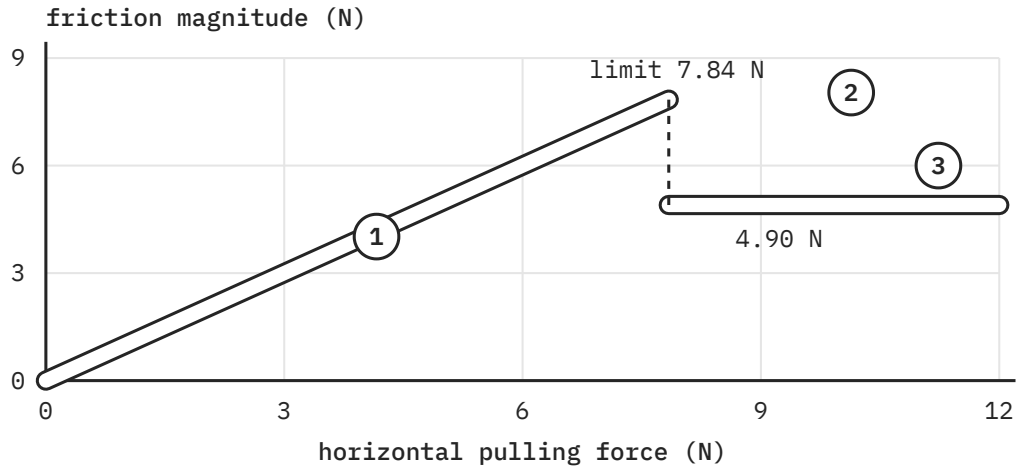
$$0 \le f_s \le \mu_s N$$

$$f_k = \mu_k N$$

$\mu$ no units
$N$ $mg$ here: level floor

## SECTION C COLOR IT · BEFORE AND AFTER SLIP

*2 kg; $\mu_s$ = .40; $\mu_k$ = .25.*




☐ **1 · ORANGE** Static friction

☐ **2 · ORANGE** Static limit: **7.84 N**

☐ **3 · ORANGE** Sliding level: **4.90 N**

## SECTION D WORKED EXAMPLE

For this box, compare a **6 N** pull while at rest with a **10 N** pull after sliding starts. Use $g$ = **9.8 m/s²**; pulls are horizontal.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $N$ = 2(9.8) = **19.6 N**<br>$\mu_s$ = 0.40<br>$\mu_k$ = 0.25 | Check $\mu_s N$.<br>Sliding: use $\mu_k N$, then net force. | $f_s$,max = **7.84 N**<br>at rest: $f_s$ = **6 N**<br>$a$ = (10 – 4.9)/2 | **0; 2.55 m/s²**<br>At rest, then sliding. |

## SECTION E THE 2026-XJ THREAD

DAY 17 OF 22 · NEW TOOL: FRICTION STATES

Surface friction is not a force that continually brakes XJ in empty space. Gravitational forces can still change its velocity, and atmospheric drag would matter in an atmosphere.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Friction changes when slipping begins

Adjustable static friction and the kinetic-friction model

If I push gently on a heavy box and it stays still, friction has matched my push. Pushing a little harder can make friction grow while the box remains at rest. That behavior shows why we cannot assign one fixed friction force to every situation. We first need to decide whether the surfaces are slipping relative to each other.

Static friction prevents relative slipping when it can. Its magnitude adjusts from zero up to a limit: $\mathbf{0 \le f_s \le \mu_s N}$. The coefficient $\mu$ is dimensionless, and the subscript s identifies the static case. The product of the coefficient and normal force gives the maximum available static friction, not necessarily the friction currently acting.

Our box has mass **2 kg** on a level surface, with a horizontal pull and no vertical acceleration. Therefore $\mathbf{N = mg = (2\ kg)(9.8\ m/s^2) = 19.6\ N}$. With $\mathbf{\mu_s = 0.40}$, the static limit is $\mathbf{f_{s,max} = (0.40)(19.6\ N) = 7.84\ N}$. We find the normal force first because another vertical force would change this limit.

Use the static-friction limit in **Section B** to check this case. Pull horizontally with **6 N** while the box is at rest. Static friction can supply 6 N in the opposite direction because that is below its 7.84 N limit. The net force is **6 N − 6 N = 0 N**, so acceleration is zero. Using 7.84 N as the actual friction here would falsely predict a backward acceleration from a gentle forward push.

Once the box slides, we use the kinetic-friction model $\mathbf{f_k = \mu_k N}$. With $\mathbf{\mu_k = 0.25}$, its magnitude is $\mathbf{f_k = (0.25)(19.6\ N) = 4.90\ N}$. For a **10 N** pull in the direction of sliding, $\mathbf{F_{net} = 10\ N - 4.90\ N = 5.10\ N}$, giving $\mathbf{a = 5.10\ N \div 2\ kg = 2.55\ m/s^2}$ forward.

Trace the rising segment in **Section C** up to the slip threshold. It shows static friction rising with the pull until the limit, followed by a lower kinetic-friction level once sliding occurs. That level is constant in our simplified model because the normal force and kinetic coefficient stay constant. Real surfaces can behave more complicatedly, but this distinction between adjusting static friction and modeled sliding friction is the starting point.

Friction opposes relative slipping, or the tendency to slip, at the contact. It does not always oppose an object's overall motion in every situation. A shoe pushing backward on the ground can receive forward static friction that helps a person walk. I want the contact and its slipping tendency identified before we choose the arrow's direction.

For our fictional asteroid, 2026-XJ, empty space does not provide a surface that continually applies sliding friction. Gravity can still change its velocity, and atmospheric drag would matter inside an atmosphere. Naming the actual interaction keeps our forecast from including a familiar force where its physical source is absent.

# Equilibrium balances components

**LEARNING TARGET** I can find a missing force when acceleration is zero.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Equilibrium** | Zero $a$: forces sum to zero on each axis. |
| **Balance** | Vertical parts support the weight. |
| **Symmetry** | Equal angles and loading: equal tensions. |
| **Rest** | $v = 0$ alone does not prove balance. |

## SECTION B KEY EQUATIONS + UNITS

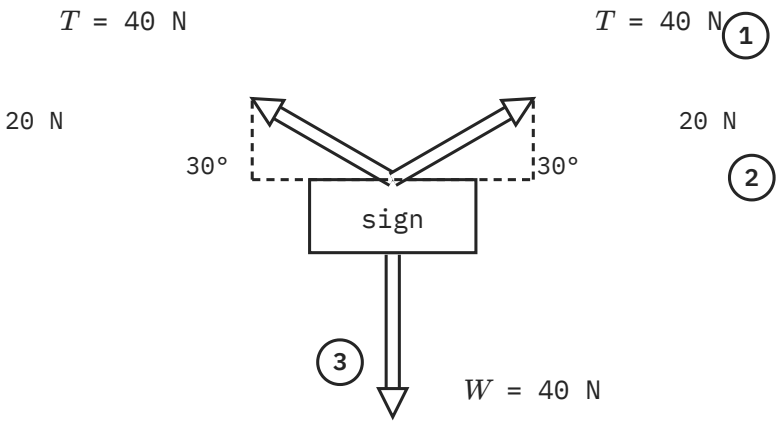
$$\Sigma F_x = \Sigma F_y = 0$$

$$2T \sin\theta = W$$

$T$ force per rope, N
$\theta$ above horizontal

## SECTION C COLOR IT · VERTICAL PARTS ADD

*40 N sign; equal 30° ropes.*




☐ **1 · ORANGE** Rope tensions

☐ **2 · ORANGE** Vertical parts: **20 N**

☐ **3 · ORANGE** Weight: **40 N**

## SECTION D WORKED EXAMPLE

A **40 N** sign hangs at rest from two equal ropes, each **30°** above horizontal. **Find the tension in each rope.**

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $W$ = **40 N**; $\theta$ = **30°**<br>equal tensions<br>$a$ = 0 | $x$ parts cancel.<br>$2T \sin$ **30°** = **40 N**. | $2T(0.5)$ = **40 N**<br>$T$ = **40 N** | **40 N per rope**<br>Each rope gives **20 N** upward. |

## SECTION E THE 2026-XJ THREAD

DAY 18 OF 22 · NEW TOOL: COMPONENT BALANCE

Zero net force would give XJ a constant velocity, not a guaranteed stop. Analysts must check each coordinate direction before treating its motion as steady.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Balance each direction, not just the force sizes

Equilibrium, angled ropes, and component sums

A hanging sign can be supported by two ropes, yet each rope may need more tension than half the sign's weight. The reason is direction. An angled rope supplies both a horizontal and a vertical force component. Only its vertical part supports the weight. Today I want us to make that geometry part of the force balance.

Equilibrium means acceleration is zero. In two dimensions, both component sums must vanish: $\Sigma F_x = 0$ and $\Sigma F_y = 0$. This can describe an object remaining at rest or moving with constant velocity. Merely observing zero velocity at one instant is not enough; an object at a turning point can be stopped momentarily while accelerating.

Our sign weighs **40 N** and remains at rest. It hangs from two ropes arranged symmetrically, each **30° above horizontal**. Equal angles and symmetric loading give equal tension magnitudes. The left rope pulls upward and left; the right rope pulls upward and right. Both are actual forces on the sign, along with Earth's downward gravitational force.

Use the angle marked in **Section C**. For each rope, the horizontal component has magnitude $T_x = T \cos 30°$. Those components point opposite ways and cancel. The vertical component of each rope is $T_y = T \sin 30°$. Both point upward, so they add. The vertical balance becomes $2T \sin 30° - 40\ \text{N} = 0$.

Since the sine of 30° is one half, $2T(0.5) = 40\ \text{N}$, giving $T = 40\ \text{N}$ in each rope. Each rope contributes $(40\ \text{N}) \sin 30° = 20\ \text{N}$ upward. The two upward components support the 40 N weight. A claim of 20 N tension per rope would confuse the vertical component with the full tension along the rope.

Check the vertical sum in **Section B** against the two vertical arrows in **Section C**. The orange diagonal arrows show the full tensions; the vertical parts show their contributions to support. We must not add each full tension and then add its components again. The component picture is another description of the same force. As with the ramp, it is a calculation tool rather than a set of extra interactions.

The angle is measured above horizontal, which is why sine gives the vertical part. If the angle were measured from vertical, cosine would give that part instead. The physical result cannot depend on how we label the angle. More nearly horizontal ropes need greater tension to supply the same vertical support, because each newton along the rope contributes less upward force. Perfectly horizontal ropes could not support this weight with any finite tension in this ideal model.

For our fictional asteroid, 2026-XJ, zero net force would mean its velocity stays constant, not that it necessarily stops. We would need to check every coordinate direction before making that claim. I want a balance calculation to include the components, their signs, and the reason equilibrium is justified, rather than relying on equal-looking arrows alone.

# Find the first wrong step and repair it

**LEARNING TARGET** I can use a picture to identify a mistake and explain a correction.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Quantity** | Name what is found: position, speed, etc. |
| **Sign** | Direction uses signs; speed is a size. |
| **Force** | Sum all forces on the same object. |
| **Check** | Use a graph or another equation. |

## SECTION B KEY EQUATIONS + UNITS

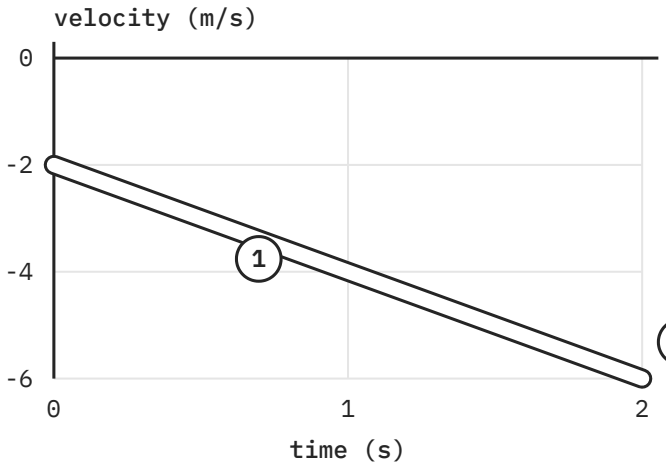
$$\Sigma F = ma$$

$$\Delta v = v_f - v_i$$

**units** N/kg = m/s²
**repair** first wrong step

## SECTION C COLOR IT · CHECK THE CHANGE IN v

*-2 to -6 m/s in 2 s.*




~~Wrong: $\Delta v = +4$~~ 3

$\Delta v = -6 - (-2)$

$= -4$ m/s

2 speed: 2 → 6 m/s

☐ **1 · PURPLE** Velocity line

☐ **2 · GREEN** Increasing speed

☐ **3 · PENCIL** Incorrect $\Delta v$

## SECTION D WORKED EXAMPLE

A **3 kg** cart has **9 N** right and **3 N** left. Someone writes $a = 9/3 = 3$ **m/s². Explain and repair the first wrong step.**

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $m$ = **3 kg**<br>right is +<br>forces: +**9 N** and −**3 N** | Use net force, not the largest force. | $\Sigma F_x = 9 - 3 = +\mathbf{6}$ **N**<br>$a_x = 6/3 = +\mathbf{2}$ **m/s²** | +**2 m/s²**<br>Check: $ma$ = **6 N**. |

## SECTION E THE 2026-XJ THREAD

DAY 19 OF 22 · NEW TOOL: REASONING CHECKS

Before the fictional XJ forecast is shared, check its signs, units, starting coordinates and force sums. Repairing the first wrong step can change every later result.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Find the first step that changes the story

Repairing force sums, signs, and interpretations

When two solutions disagree, finding a different final number is not enough. We need to locate the first step where the reasoning stops matching the physical situation. A later calculation can be perfectly executed and still carry forward an earlier mistake. Today I want us to explain the repair, not just replace an answer.

Start by naming the quantity being found and the reference frame. Position, displacement, velocity, speed, and acceleration answer different questions. Their units also differ. A number can have the right sign for position and the wrong interpretation for velocity. Keeping the quantity's name beside the number is a simple way to make the reasoning visible.

In the force example in **Section D**, a **3 kg cart** experiences **9 N right** and **3 N left**. Someone divides the larger force by mass and reports 3 m/s². The first error is selecting the force, not performing the division. Newton's second law uses the net force on the object. With right positive, $\Sigma F_x = 9\ \text{N} - 3\ \text{N} = +6\ \text{N}$.

The corrected acceleration is $a_x = +6\ \text{N} \div 3\ \text{kg} = +2\ \text{m/s}^2$. Check it by multiplying mass and acceleration: $(3\ \text{kg})(2\ \text{m/s}^2) = 6\ \text{N}$, which matches the force sum. The unit check works because a newton is a kilogram-meter per second squared, so newtons divided by kilograms give m/s². The force diagram also shows a rightward imbalance, matching the acceleration sign.

Trace the line from left to right in **Section C**. Decide whether its change in height is positive or negative. Velocity changes from $-2\ \text{m/s}$ to $-6\ \text{m/s}$ in 2 s. Final minus initial gives $\Delta v = -6\ \text{m/s} - (-2\ \text{m/s}) = -4\ \text{m/s}$. Subtracting a negative adds its positive counterpart; it does not make the final result automatically positive. Therefore $a_{\text{avg}} = -4\ \text{m/s} \div 2\ \text{s} = -2\ \text{m/s}^2$.

That negative acceleration goes with increasing speed in this interval. The object moves in the negative direction throughout, while speed grows from 2 m/s to 6 m/s. The downward slope of the velocity graph means velocity becomes more negative. It does not mean the object is slowing. Purple identifies signed velocity, and green marks the increasing speed.

A useful check should examine the reasoning from another angle. Compare a calculation with a graph, inspect whether all forces were included, or test whether the units and limiting behavior make sense. Repeating the same mistaken substitution in another equation may reproduce the same error. If one step changes, follow that correction through every result that depends on it.

For our fictional asteroid, 2026-XJ, an incorrect starting coordinate, force sum, or sign can shift an entire forecast. Before sharing a prediction, we should trace it back through the measurements, chosen frame, and assumptions. An explanation of the first wrong step makes the correction understandable and shows why the revised conclusion deserves more confidence.

# Connect forces to velocity and position

**LEARNING TARGET** I can connect forces, velocity change, and final position.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Chain** | Net force → $a$ → velocity and position. |
| **Interval** | Keep one start and end time throughout. |
| **Reversal** | $v$ can cross zero while $a$ keeps its sign. |
| **Area** | Add positive and negative pieces of $\Delta x$. |

## SECTION B KEY EQUATIONS + UNITS

$$a = \Sigma F \div m$$

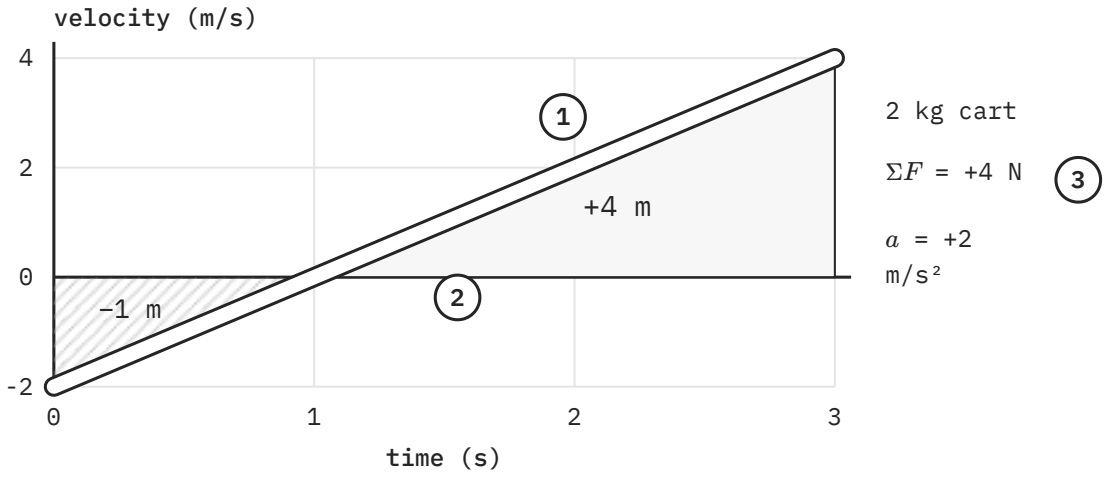
$$v_f = v_i + at$$

$x_f$ $x_0 + v_i t + \frac{1}{2}at^2$

`model` constant $a$

## SECTION C COLOR IT · FORCE TO POSITION

*Velocity crosses zero at **1 s**.*




☐ **1 · PURPLE** Velocity line

☐ **2 · BLUE** Signed areas

☐ **3 · ORANGE** Net force: **+4 N**

## SECTION D WORKED EXAMPLE

A **2 kg** cart starts at **1 m** with velocity **−2 m/s**. A constant **+4 N** net force acts for **3 s**. **Find final velocity and position.**

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $m$ = **2 kg**; $\Sigma F$ = **+4 N**<br>$x_0$ = **1 m**; $v_i$ = **−2 m/s**<br>$t$ = **3 s** | Find $a$ from force.<br>Then steady-a equations. | $a$ = 4/2 = **+2 m/s²**<br>$v_f$ = −2 + 2(3)<br>$x_f$ = 1 − 6 + 9 | **+4 m/s; +4 m**<br>$\Delta x$ = **+3 m**; distance = **5 m**. |

## SECTION E THE 2026-XJ THREAD

DAY 20 OF 22 · NEW TOOL: THE FULL CHAIN

XJ's forecast needs its present motion and the forces changing it. Use one reference frame through the entire calculation.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Follow the chain from force to position

Constant net force, velocity reversal, and signed displacement

We now have the pieces needed to predict motion from forces. The net force gives acceleration. Acceleration changes velocity. Velocity over time changes position. I want each link to use the same object, reference frame, and time interval. That consistency matters especially when an object reverses direction during the calculation.

Start with the given values in **Section D**. Our **2 kg cart** starts at **1 m** with velocity **−2 m/s**. A constant net force of **+4 N** acts for **3 s**. Take right as positive. The first step is $a = \Sigma F \div m = 4\ \text{N} \div 2\ \text{kg} = +2\ \text{m/s}^2$. Because force and mass stay constant, acceleration stays constant over the interval.

Next find final velocity: $v_f = v_i + at = -2\ \text{m/s} + (2\ \text{m/s}^2)(3\ \text{s}) = +4\ \text{m/s}$. The cart began moving left and finishes moving right. Its velocity passes through zero at $t = (0 - (-2))\ \text{m/s} \div (2\ \text{m/s}^2) = 1\ \text{s}$. Positive acceleration first slows the negative motion, then builds positive motion.

The displacement equation covers the entire constant-acceleration interval, including the turn: $\Delta x = v_i t + \tfrac{1}{2}at^2$. Substitution gives $\Delta x = (-2\ \text{m/s})(3\ \text{s}) + \tfrac{1}{2}(2\ \text{m/s}^2)(3\ \text{s})^2 = -6\ \text{m} + 9\ \text{m} = +3\ \text{m}$. Finally add the starting coordinate: $x_f = 1\ \text{m} + 3\ \text{m} = 4\ \text{m}$. Displacement and final position are different quantities.

Compare the areas below and above the time axis in **Section C**. Their signed sum explains why displacement is only 3 m. During the first second, the velocity line lies below zero. Its triangular area is $-\tfrac{1}{2}(1\ \text{s})(2\ \text{m/s}) = -1\ \text{m}$. During the next two seconds, the line lies above zero, giving $+\tfrac{1}{2}(2\ \text{s})(4\ \text{m/s}) = +4\ \text{m}$. Adding the signed areas gives the same +3 m displacement. The blue regions track position change; the purple line tracks velocity.

Distance uses the sizes of those two regions instead: **distance = 1 m + 4 m = 5 m**. The cart travels from its starting coordinate of 1 m leftward to 0 m, then rightward to 4 m. That route makes the distinction concrete. A positive final displacement does not mean every part of the trip was in the positive direction.

We can check the chain in several ways. The net force and acceleration point the same way. Velocity changes by 6 m/s over three seconds at 2 m/s². The graph's signed areas agree with the displacement equation. Adding the initial position gives the correct final coordinate. If the force changed during the trip, this single constant-acceleration calculation would need to be divided into suitable intervals or replaced with a more detailed model.

For our fictional asteroid, 2026-XJ, a forecast needs both its present motion and the forces changing that motion. Starting with force alone leaves out initial velocity and position. Connecting all three ideas gives us a prediction we can explain, check, and revise when new evidence arrives.

# Plan a prediction in two dimensions

**LEARNING TARGET** I can plan a prediction using evidence and fixed axes.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Frame** | Keep the origin and axes fixed. |
| **Components** | Signed $x$ and $y$ parts; calculate separately. |
| **Units** | Make all units match before substituting. |
| **Limit** | State assumptions, such as steady velocity. |

## SECTION B KEY EQUATIONS + UNITS

$$x_f = x_0 + v_x t$$

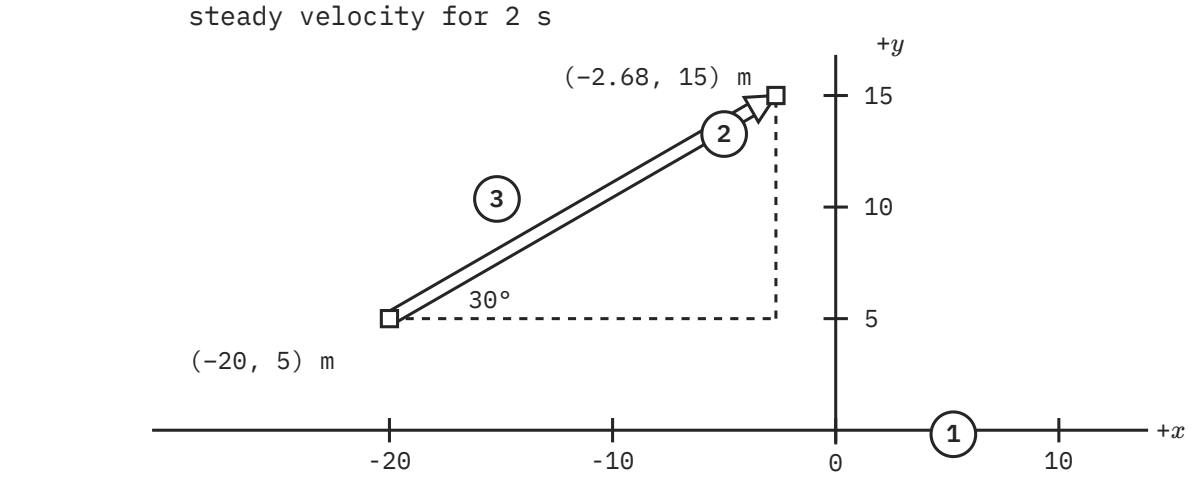
$$y_f = y_0 + v_y t$$

$v_x$ $v \cos\theta$

$v_y$ $v \sin\theta$; $\theta$ from $+x$

## SECTION C COLOR IT · START + CHANGE

*Probe example; fixed axes.*




☐ **1 · PENCIL** Origin and axes

☐ **2 · BLUE** Start and end positions

☐ **3 · BLUE** Displacement direction

## SECTION D WORKED EXAMPLE

A probe starts at (−20, 5) m and moves at **10 m/s**, **30°** from $+x$ toward $+y$, for **2 s**. **Find its new coordinates.**

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $x_0$ = **−20 m**; $y_0$ = **5 m**<br>$v$ = **10 m/s**; $\theta$ = **30°**<br>$t$ = **2 s**; steady $v$ | Resolve $v$:<br>$v_x$ = **8.66 m/s**<br>$v_y$ = **5 m/s** | $x_f$ = −20 + 8.66(2)<br>$y_f$ = 5 + 5(2) | **(−2.68, 15) m**<br>Final $x$ is still negative. |

## SECTION E THE 2026-XJ THREAD

DAY 21 OF 22 · NEW TOOL: A TWO-AXIS PLAN

For fictional XJ, name the frame, units and motion assumptions. Fixed Earth and steady velocity are classroom simplifications.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Build a prediction one coordinate at a time

A two-dimensional plan with fixed axes and clear assumptions

A two-dimensional prediction can look complicated because the motion points at an angle. The solution becomes manageable when we keep one fixed frame and calculate each coordinate separately. I want the plan to show what was measured, what is being assumed, and how the result follows. Our probe example gives us a way to practice that reasoning.

Locate the probe's starting point in **Section C**. It starts at **(−20, 5) m** and moves steadily at **10 m/s**, directed **30° from the positive horizontal axis toward the positive vertical axis**. We want its position after **2 s**. The coordinates describe where it starts. The speed and angle describe its velocity. They do not describe the same thing, and the negative starting coordinate does not mean negative horizontal motion.

Resolve the velocity along the fixed axes. Because the angle begins at the positive horizontal axis, $v_x = v \cos 30° = (10\ \text{m/s}) \cos 30° \approx 8.66\ \text{m/s}$. The vertical component is $v_y = v \sin 30° = 5.00\ \text{m/s}$. Both are positive. Using 10 m/s independently along each axis would invent a larger velocity than the one we were given.

Use the two steady-velocity equations in **Section B**, one for each coordinate: $x_f = x_0 + v_x t$ and $y_f = y_0 + v_y t$. Both calculations use the same two seconds. Horizontally, $x_f = -20\ \text{m} + (8.66\ \text{m/s})(2\ \text{s}) \approx -2.68\ \text{m}$. Vertically, $y_f = 5\ \text{m} + (5\ \text{m/s})(2\ \text{s}) = 15\ \text{m}$.

The predicted coordinates are therefore **(−2.68, 15) m**, using rounded horizontal components. The probe has moved rightward, yet remains left of the vertical axis. That is consistent: starting at −20 m and moving 17.32 m right does not quite reach zero. The blue start and end points on the notes page should tell the same story as those signed calculations.

Check the displacement components: $\Delta x \approx +17.32\ \text{m}$ and $\Delta y = +10\ \text{m}$. Their combined magnitude is approximately 20 m, matching a speed of 10 m/s maintained for two seconds. The displacement points in the same direction as the steady velocity. Agreement between the component calculation, diagram, and total travel is stronger evidence than a calculator result alone.

A complete plan also checks that all units match before substitution and that the chosen model fits the interval. If velocity changes, the two steady-velocity equations cannot describe the whole trip as written. If the starting measurements have uncertainty, the final coordinates should carry appropriate uncertainty too. More digits in the arithmetic do not create more precise measurements.

For our fictional asteroid, 2026-XJ, treating Earth as fixed and velocity as steady are classroom simplifications. Real orbital predictions must account for changing gravitational effects and Earth's motion. Within our model, we can still make a defensible claim by naming the axes, future time, assumptions, and predicted coordinates, then asking whether both objects occupy the same place at the same time.

# Defend and revise a forecast

**LEARNING TARGET** I can defend a prediction, state limits and revise it.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Prediction** | Result, units, time, direction and range. |
| **Evidence** | A measurement or calculation supporting a step. |
| **Limit** | An uncertain input or missing effect. |
| **Revision** | Name the change and show its effect. |

## SECTION B KEY EQUATIONS + UNITS

$$x_f = x_0 + vt$$

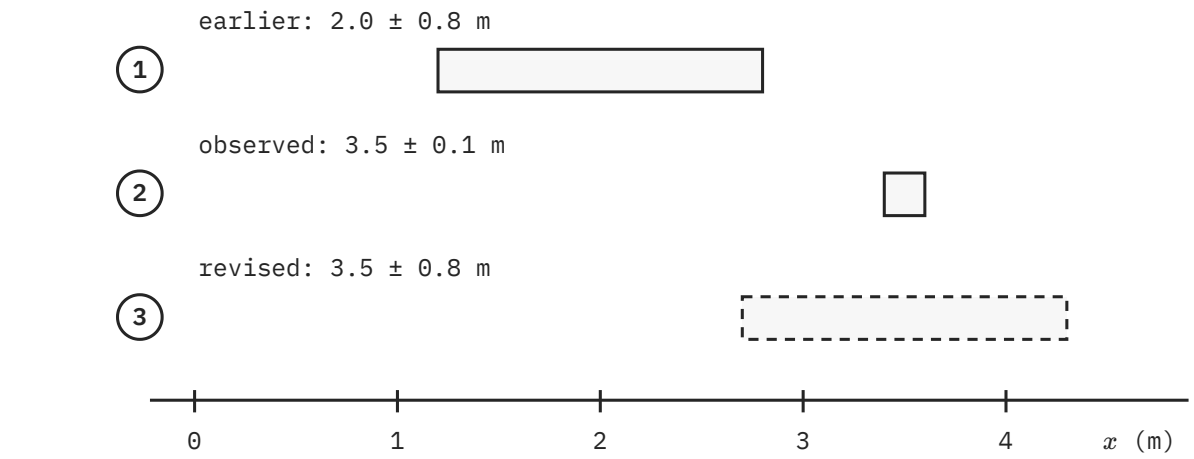
$$\delta x_f = \delta x_0 + t\delta v$$

$t$ exact time

**bounds** not probabilities

## SECTION C COLOR IT · COMPARE THE RANGES

*Illustrative data; same axis.*




**1 · BLUE** Earlier forecast

**2 · BLUE** Observed range

**3 · BLUE** Revised forecast

## SECTION D WORKED EXAMPLE

Keep $x_0$ = **8 ± 0.5 m** and exact $t$ = **3 s**. A new measurement changes $v$ from −2.0 to **−1.5 ± 0.1 m/s**. **Update the forecast.**

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $x_0$ = **8 ± 0.5 m**<br>$v$ = **−1.5 ± 0.1 m/s**<br>exact $t$ = **3 s** | Update center and conservative bound. | $x_f$ = 8 − 1.5(3)<br>$\delta x$ = .5 + 3(.1)<br>= **0.8 m** | **2.7 to 4.3 m**<br>Center shifts +**1.5 m**. |

## SECTION E THE 2026-XJ THREAD

DAY 22 OF 22 · NEW TOOL: A REVISED FORECAST

Show the evidence behind XJ's forecast and what changed after review. Separate the physics from cost and feasibility decisions.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# A good forecast can change with the evidence

Defending a claim, comparing ranges, and explaining revision

A forecast is useful when someone else can follow the reasoning behind it. Giving a location without a time, reference frame, or uncertainty leaves too much unsaid. Today we bring those pieces together and examine what happens when a new measurement changes the result. Revising a prediction is part of using evidence responsibly.

Begin with the earlier constant-velocity model: $x_f = x_0 + vt$. The starting position was **8.0 ± 0.5 m**, velocity was **−2.0 ± 0.1 m/s**, and elapsed time was exactly **3 s** in the model. The central prediction was **8.0 m + (−2.0 m/s)(3 s) = 2.0 m**. Its conservative uncertainty bound was **0.5 m + (3 s)(0.1 m/s) = 0.8 m**, giving an interval from **1.2 m to 2.8 m**.

Follow the revised forecast in **Section D**. A new measurement changes the velocity estimate to **−1.5 ± 0.1 m/s**. Keep the same starting-position estimate and exact elapsed time. The revised center is $x_f$ **= 8.0 m + (−1.5 m/s)(3 s) = 3.5 m**. The new velocity is less negative, so the predicted decrease in position is smaller. That sign check explains why the final position moves upward on the number line.

The center shifts by **(0.5 m/s)(3 s) = +1.5 m**. The uncertainty bound stays $\delta x_f = \delta x_0 + t\delta v$ **= 0.8 m**, because the input bounds did not change. We now report $x_f$ **= 3.5 ± 0.8 m**, or **2.7 m to 4.3 m**. The full interval is still 1.6 m wide. A changed center does not automatically mean a changed uncertainty.

Compare the three ranges on the same axis in **Section C**. The middle range shows an illustrative observed position of **3.5 ± 0.1 m**, an interval from 3.4 m to 3.6 m. That observed interval does not overlap the earlier forecast, but lies within the revised forecast. The new prediction is consistent with this observation under the stated bounds. Agreement supports the model for this comparison; it does not prove that every assumption will remain valid at later times.

Keep the symbols distinct. Capital delta, $\Delta$, means a change in a quantity. Lowercase delta, $\delta$, labels an uncertainty bound here. We are adding the largest allowed contributions, not calculating a standard deviation or an impact probability. The range also does not automatically cover an omitted effect such as acceleration. A model error and an uncertain measurement are different reasons a prediction may need revision.

I want a defense of the forecast to connect four things: the claim, the evidence, the reasoning, and the limits. State the predicted position and time in a named frame; identify the measurements; explain the equation and assumptions; and show the allowed range. When revising, identify which input or assumption changed and trace its effect through the calculation.

For our fictional asteroid, 2026-XJ, this is the standard for reporting our conclusion. We can compare forecasts with observations and update them openly. The physics tells us what motion the model predicts; decisions about cost, feasibility, or a response require additional information. A clear forecast makes that later discussion better informed without claiming more certainty than the evidence provides.